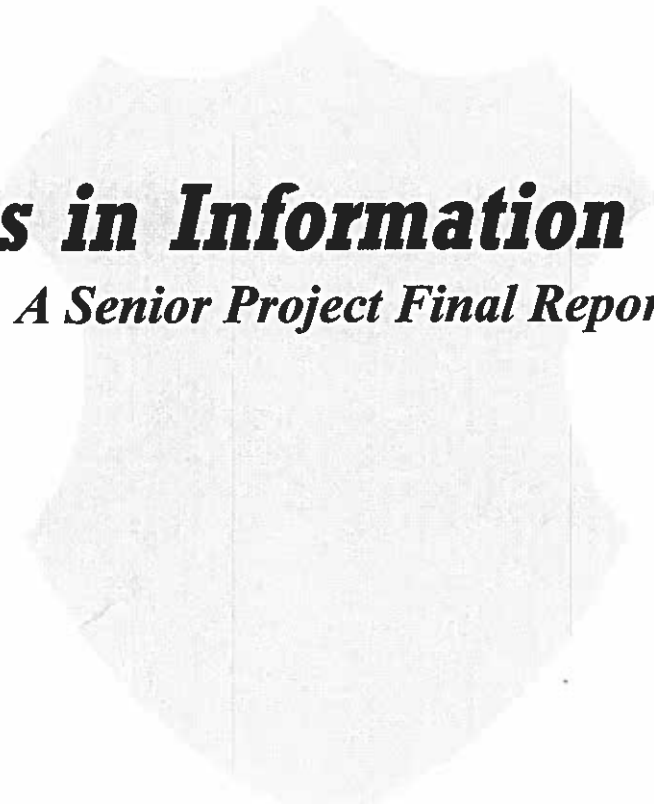




Lessons in Information Quality:

A Senior Project Final Report

by

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Dr. Valerie Sessions came to Charleston Southern with an ongoing project in data quality, working with smaller law enforcement agencies to clean up the data in their databases. Her research interests lie in developing tools to detect and clean bad data with less human interaction than currently available and more accuracy. She applied for and received a small grant to prove that good, easy to use tools could be constructed on a five digit budget, instead of a six or seven digit budget and that those tools could be effective in small law enforcement agencies of 10 or fewer officers. The project would be conducted in parallel with the Criminal Justice department, with two full professors: Dr.'s Valerie Sessions and Jaqueline Fish and one instructor, Officer Linn Pitts, four research assistants: Steven Garbeil and Rebecca Randall from COINS, and Jasmine Robinson and Jamie Simkulet from Criminal Justice.

The principles of data quality management are simple and can be applied to low volumes of data, as well as huge relational databases. This is in comparison to the project that other schools and research institutes are handling with expensive tools and huge enforcement agencies. Our (Charleston Southern University) part of the study would be to have a 'low tech' solution, and a high tech solution as a demonstration toolset, accessed from a web page, with hardware and software support for off campus access.

The 'low tech' solution would be a training style video, emphasizing the dangers of ignoring data quality problems, dealing with mistakes in entry and analysis, and pointing the way to some tools to fix the issues. This is an ongoing project in four or more parts, in which this is the first. The first part is designed to be more in the nature of an advertisement, then an actual working setup.

Steven was asked to participate because he had yet to submit a topic request for senior project, and secondly; Steven is a professional programmer, and was thrilled to be making a small movie instead of a code-based project. Rebecca, also in need of a Senior Project, was offered the task of doing the 'high tech' solution, a web site with movies and pages, with voice over, and she was also pleased at the opportunity to participate in grant work.

The project is a boon for the school and the timing is excellent, considering the economic climate.

Introduction

This report outlines the Computer and Information Systems department's part of a thirty-week grant from the National Institute of Justice (NIJ) awarded based on an application by Dr. Sessions. The Statement of Work (SOW) is included in Appendix B. This grant is an effort by Dr. Sessions to continue her valuable work by creating an example or demonstration of analysis tools and the reasons they should be used. By applying rigorous mathematical analysis of data sets, it should be possible to greatly reduce the incidents of hours lost due to bad data. The continuation of the work to set the next stage is actually using the processes developed by Dr. Sessions. Before this can happen, small Law Enforcement agencies must be convinced that Information Quality Assessments are not a waste of their time.

Small Agencies are strapped for time and man-power and thus are not easily swayed to use these tools. The idea of showing them how bad data can effectively waste days of time in the form of training type videos and web pages

This report is in three parts, the low tech video, the high tech video, and a presentation, or web delivery portion. The first and second parts detail the process of creating a 'low tech' movie and 'high tech' movie for web delivery. The third part details the creation of the web page framework and delivery of the movies to be viewed off campus.

Part One: The 'Low Tech' Movie

Making a movie, even a small short or two small shorts and some interlinking narration turns out to be an extremely arduous undertaking, involving many steps and some expertise not normally found in ordinary computer literati. Several major universities have a computer science emphasis on video. Tools are required that are not in the normal repertoire of the computer 'hacker'. The skills include minimal sketching, storyboarding, plot dynamics and layout, creative writing, a little video knowledge for scripting conversions and formatting.

The first step is a topic. Not so easy for this project, the general outline was fixed, but the topic or theme was not. The statement of work outlined the basics of the presentation, but the medium and format was left up to the workers. To decide upon a topic, we held a session with Dr. Sessions and she laid out her ideas for the final product. This gave a framework to overlay a storyline. In fact she related an anecdote, from personal experience

that gave birth to the third candidate script, and then became the second video after some creative dialogue.

The process creates quite a bit of paperwork: manuscripts, storyboards, scripts, prop lists, casts. This sequence is called pre-production, and there are a number of programs that are set up to handle all this 'stuff' and keep the details consistent. This project used Celtx, a free studio pre-production tool by Celtx, an animation company, to handle pre-production of commercials and the like, which they consult for a fee. The tool is free of charge for home and non-profit use, and other use as laid out in the license. CSU meets the terms for free use and thus uses a very nice tool gratis.

Starting the journey to a motion picture video, a cartoon animation, or a three dimensional graphic sequence, one starts out in the same place the movie studios such as Disney Animation, Pixar, DreamWorks, or ILM, with a storyboard. A storyboard is the easiest, if not the only, way to create and sequence a storyline.

Storyboarding, as it is known today was created at Disney Picture Studios, for use on Snow White (Holt 1956). It consists of a bunch of squares of paper with thumbnail sketches tacked up on a corkboard, pasted to a blackboard, or yellow stickied to a whiteboard etc. Obviously Walt Disney didn't use stickies, but their format is similar. This project used a wheel book for portability, and scanned them into Celtx for publication (email, html, word processing etc.) This allows sequencing to change merely with 'drag-n-drop' and to do a sequence session by conference or email.

We created three storyboard sequences, pencil sketches in block frame format, plus several

more sequences for the narration and the IQ tools scenes and then submitted them first to Dr. Sessions, and then to the Contracting Officer Representative (COR) at SPAWAR for approval, receiving approval to go ahead.

The next step is a script outline, or plot synopsis. This is written up in simple give and take format, as a simple storyline. After a read through of the simple dialogue, which we did, Dr. Sessions and Dr. Fish gave their approval and we went on to create the full manuscript for all three scenarios. The second scenario wasn't produced, but we did the full story because it was humorous.

Creating a script using Celtx is much easier than using a word-processing tool such as MS Word. It handles the prop list and shot sequence automatically. With the tool one can pull up a set of shots that use the same scenery all throughout the film, and allows offline collaboration and seamless archival of the pre-production stage.

A movie or animation script is not quite the same as a play script even though the formats are similar. An animation script will generally have more margin notes and rendering information, backtracking data and some other computer specific detail.

A training film is required to have a certain 'look' and the script helps to capture this. The script was distributed to the advisors, revised and commented, and given out to the actors. Actors are people, and during this particular project were less than reliable. The narrator did not memorize his lines so we had to create a teleprompter during the shoot. The actors did not learn their lines, and they ended up switching parts. The shoot was delayed by an entire month and so a video service was retained to help with the shooting and editing.

The shoot was done in sequences of two days with a morning and afternoon shoot for each. The cameraman, Mark Kennedy, used a Sony Beta Cam, a very expensive professional quality video camera. It saves the video and audio in Beta format with digital markers. The scenes were transcribed using a professional quality video deck onto DVD with a frame grabber. This is a big step beyond iMovie. The editing took place over four days, and was mostly accomplished using Adobe Premiere Professional CS3, with some help from sound editors and Adobe Photoshop. Video editing requires a multi-track video session recorder, and since about fifteen years ago these editors have been computers. Digitizing the film and editing real time saves literally thousands upon thousands of dollars of labor per film.

Following editing, the movie had to be converted for use in various delivery systems. The web requires a small, medium, and large format for stepped bandwidth, and this is an advantage to the normal strategy. Software must be written to package the raw video and compress it according to one or another codes, depending on the end user consumption. These programs are now in the form of scripts (used to be coded directly) to access codecs (compression-decompression libraries.)

Part Two: The 'High Tech' Movie

Another video was created as part of the project. This video was designed to show how multiple agencies come together to use data. This video would be used to inform Law Enforcement of two powerful tools that are available: GJXML/NIEM standards and the Self-Assessment Tool. The video, in addition to briefly introducing the tools, shows Law

Enforcement how the information quality of the data being shared between different agencies would be improved by the use of these tools.

Every step involved in creating the video, had to take into consideration that it would ultimately be place on the web for others to view. This meant that the file size would be playing a big role in programs or formats would be used. The first step was to do some research on the GJXML/NIEM standards, the Self-Assessment Tool and on information quality was needed. Such research would allow us to emphasize the benefits of using the tools to improve the information quality. Once the research was done, the next step was to get a storyboard done on the video. This was done using Microsoft PowerPoint. Basically the scenario is a multistate shoplifting ring where multiple agencies come together through an online database to try to solve the crimes.

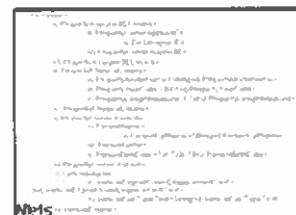
Information Quality: Brief introduction to the importance of IQ and how it is beneficial to all agencies in performing their duties, even as a team.



Narrator speaking

GJXML: Introduce GJXML, what it is and how it works

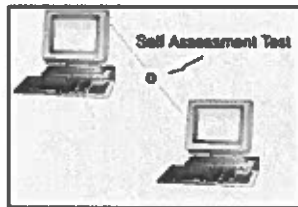
GJXML was developed by the NIJ and partnering agencies and has had much success in allowing LEAS with different systems to exchange data. This has facilitated new levels of information sharing fusion and has lead to many advances in the CJ field



Narrator speaking

Self Assessment:

despite using GJXML, nothing guarantees the quality of the information exchanged. To increase the information quality a self assessment test should be taken



With new standards to facilitate information sharing, data is passed among LEAs and partners more efficiently. In order to make sure that this data sharing is effective and successful, we must ensure that the data is accurate, timely, and of overall high quality.

Narrator speaking

Scenario:

Multi-State shoplifting ring in major retail stores.



To illustrate how the GJXML and a Self-Assessment tools enhance the information quality shared between different Law Enforcement Agency, a multi-State shoplifting ring scenario will be presented.

Narrator speaking

Scenario:

Multi-State shoplifting ring.

Major retailers through out the States of Nevada, California and Texas have felt a strong increase in shoplifting activities.



Narrator speaking

Scenario:

Retailers and LE are able to share information through an online database.

These retailers have come together with Law Enforcement (LE) through a national online database to track down the criminals. Now LE and other retailers have almost real-time access to crime information as they occur.



Narrator speaking

Scenario:

Major retailers in different states are burglarized.

A retail store in California has been burglarized and the criminals have taken thousands of dollars in goods. As soon as the incident occurs the retail store enters the details into the database and LE is working the leads.



Hours later the Las Vegas Metropolitan police seize a van full of items from shoplifting episodes all over the southwest.



Retail stores in Texas are burglarized and the criminals are caught at the border attempting to cross over with the goods.

The burglaries all seem to be linked to the same shoplifting ring.

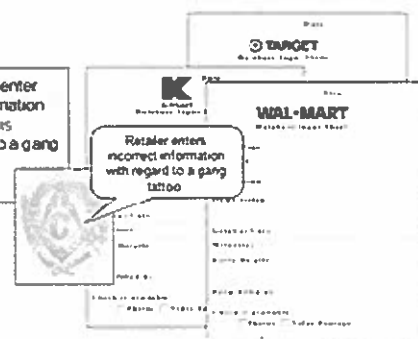


Narrator speaking

Scenario:

each of the retailers report their information into the database to share with other retailers and LE agencies.

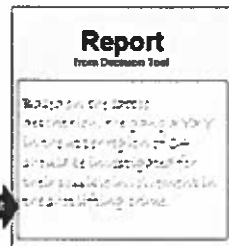
The retailers and LEAs enter their data, but the information from the retailers in CA is incorrect with regards to a gang tattoo marking.



Narrator speaking

Scenario: All the tips are entered into a decision making tool.

All the information from the different retailers and LE agencies are entered into a decision making tool.

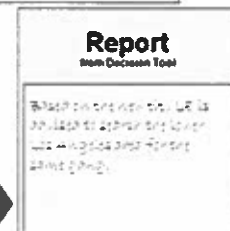


Narrator speaking

Scenario: Another LE agency provides a new tip regarding the gang.

A new tip is received from another LE agency where the man vehicle in CA's a white Chevy with Nevada license plate 555 PPP

This tip is entered into the Decision Tool



Narrator speaking

Scenario: All the tips are revised.

LE revise the information they have gather thus far. They begin to notice that if they take out the SMT (tattoo) information, a new search area is to be focused on



Narrator speaking

Scenario: LE in the correct jurisdiction is alerted.

After reviewing the information and taking out the erroneous piece of information, the correct LE agency is alerted and the suspect is caught



Narrator speaking

Scenario: Benefits of using a self assessment tool.

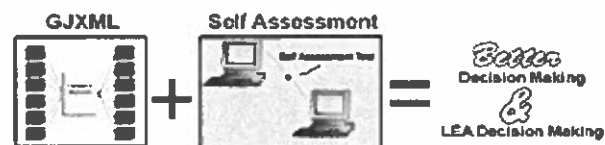
As seen from the previous scenario, LE was focusing on the wrong area due to an erroneous SMT tip. If the information provided had been processed through a self assessment tool (SAT), LE would have been on the criminals track sooner



The higher the score, the more valuable the information is

Narrator speaking

Recap: recap the benefit of using GJXML along with the self assessment tool to improve information quality



Narrator speaking

Once the storyboard was complete it was time to get the narration. The narration plays

throughout the whole video explaining what's happening at each moment.

Law Enforcement agencies across the nation gather thousands of pieces of information daily. The need to share this information with other agencies is inevitable. But what about the quality of the information being shared? How reliable is it? Does it have the same meaning for one agency as it does for another?

The Global Justice Extensible Mark-up language and National Information Exchange Model are XML standards that were developed by the National Institute of Justice and partnering agencies. They were designed for the exchange of criminal justice information. This has facilitated new levels of information sharing between different agencies in a timely manner. This common standard eliminates the need for each agency to create its own exchange standards. With all systems using the same common terms, NIEM permits access from numerous sources and reuse in numerous applications.

While using a common standard increases the timeliness and ability of agencies to share data, there is no measure of the quality of the information being exchanged. To increase the information quality and usability of the information, a self assessment tool is needed. This ensures that the data being shared is accurate, timely, and of overall high quality.

To illustrate how the GJXML/NIEM and the Self-Assessment tool enhance the information quality shared between different Law Enforcement agencies we will present a multi-State shoplifting scenario.

Shoplifting is a nationwide crime. Major retailers through out the States of Nevada, California and Texas have felt a strong increase in shoplifting activities. A retail store in California is burglarized and the criminals have taken thousands of dollars in goods. Hours later the Las Vegas Metropolitan police seize a van full of items from shoplifting episodes all over the southwest. The retail stores in Texas are also burglarized and the criminals are caught at the border attempting to cross over with the goods.

To track down the criminals, the retail stores and Law Enforcement have come together through a national online database. Each of them has entered their details into the database allowing Law Enforcement and other retailers to have almost real-time access to crime information as they occur. But the information entered into the database from the retailers in California is incorrect with regards to a gang tattoo marking. Law Enforcement believes these burglaries are linked to the same shoplifting ring.

Once all the information is gathered from the retailers and Law Enforcement agencies, it's entered into a decision making tool.

Based on the tattoo description, the decision making tool indicates that the gang known as Las Zetas in the upper region of California should be investigated for their possible involvement in the shoplifting crime.

Days later a new tip is received from another Law Enforcement agency where the main vehicle in California is a white Chevy with Nevada license plate number 555-PPP. This new tip is entered into the decision making tool and the new report advises Law Enforcement to search the lower Los Angeles area for the same gang.

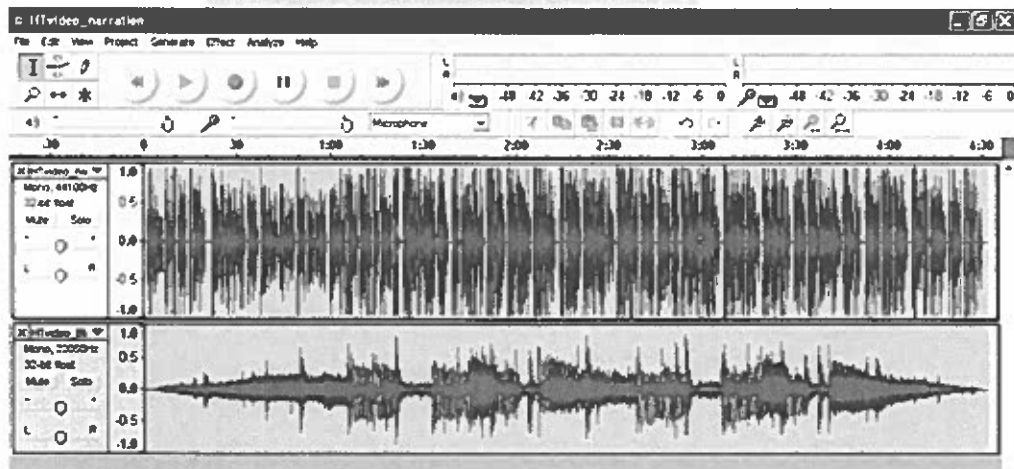
Law Enforcement begins to revise the information they have gathered thus far. They begin to notice that if they take out the SMT data, the decision making tool will advise them to search the Houston area where the leader of the Shoplifting Crime ring resides. With the new report, the correct Law Enforcement agency is alerted and the suspect is apprehended.

The erroneous piece of information was leading Law Enforcement to focus on the wrong area. Had the information been assessed through a self assessment tool, Law Enforcement would have been able to track down and catch the suspect sooner.

The GJXML/ NIEM standards and Self Assessment tools are two powerful tools developed to improve information quality for Law

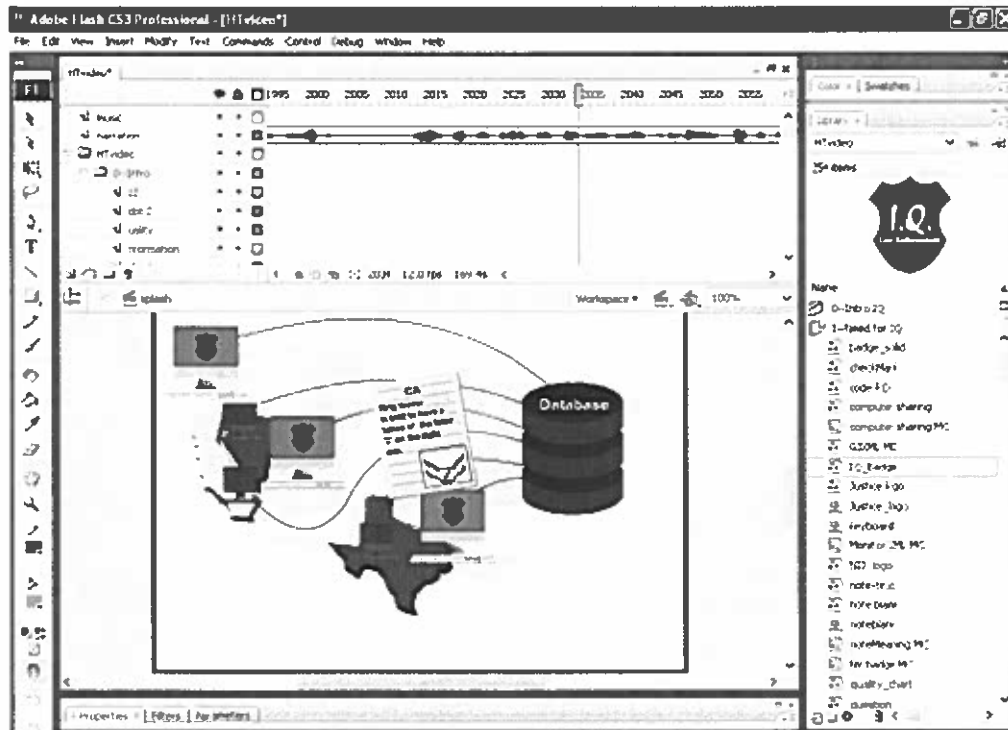
Enforcement agencies. Through their use, better decision making within Law Enforcement is possible. Through their use time, money and, more importantly, lives can be saved!

Next step was to get the audio recorded for the video. Dr Colman, the vice president of academic affairs, was kind enough to allow us to use his voice. The software used for the recording and editing the voice was Audacity. After the recording session, there were a total of eight audio files that were left to edit and piece together. Through the use of audacity, the audio files were pieced together and any external noises were removed. The final audio file was saved as an MP3 file because the file size would be compressed enough to be able to use it on the web. MP3 files are known to reduce the data needed to represent the audio while still keeping the sound good enough as the original.



The final and most important step to the creation of the video was making the animation. For the video, the choice of software was Adobe Flash CS3. Once again Flash was chosen because it allows for a small file size. In Flash, when you creating a graphic through a

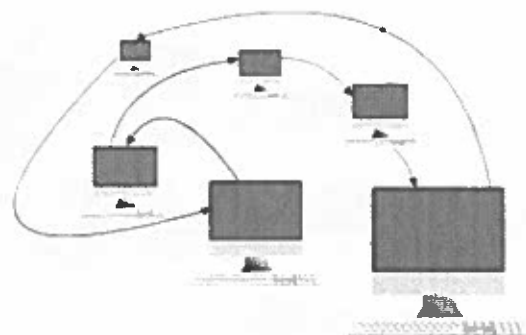
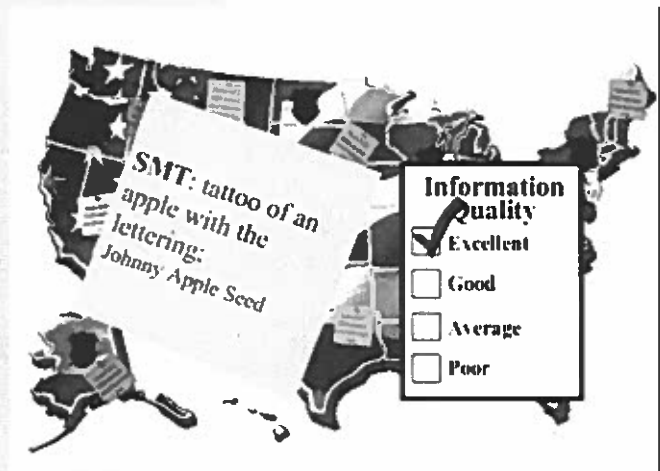
symbol, it does not matter if you use it one time or one thousand times, the file size remains the same as long as you use the same symbol each time.

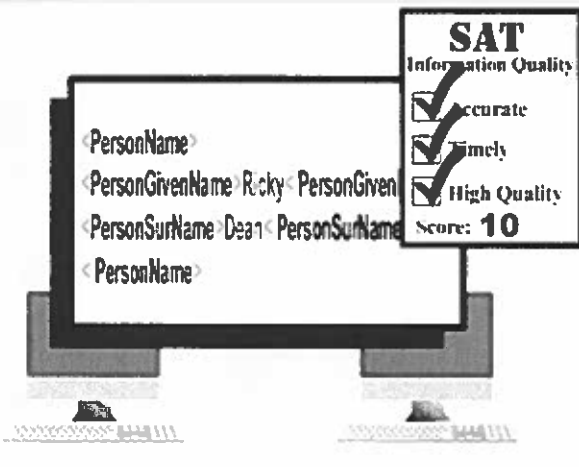


The most difficult piece for creating the video was coming up with ideas to represent the narration. Once those came to mind, the rest was not that difficult. A technical difficulty that arose during the creation of the video was synchronizing the video with the audio. The problem was that the audio needed to start at a certain frame and maintain the same speed as the animation. The audio was initially set to start. Setting the sync option to start did exactly as it was suppose to but it did not force the audio to keep up with the animation. So when it was uploaded to the server and played, the audio would go faster than the animation. After multiple attempts in moving the animation around to match the audio and not being successful, it was time to search for the reason of the synchronization problem. It all came down to not setting the audio to stream. Stream forces the audio to

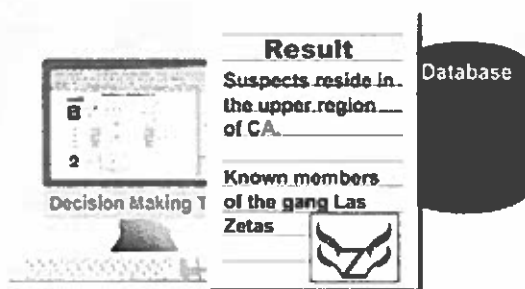
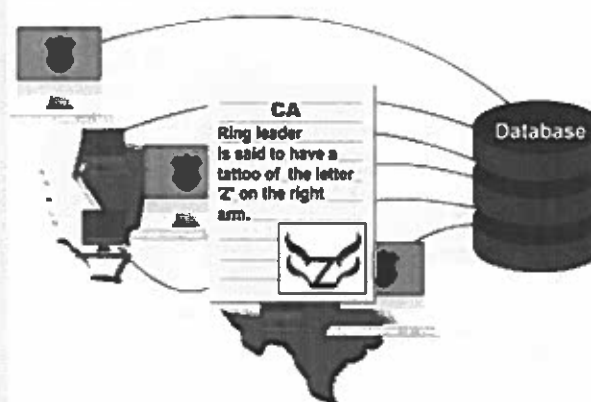
keep pace with the animation. If ones computer cannot draw each frame, Flash will skip the necessary number of frames in order to keep up with the sound. This eventually fixed the problem and finally the audio and the animation were matching perfectly.

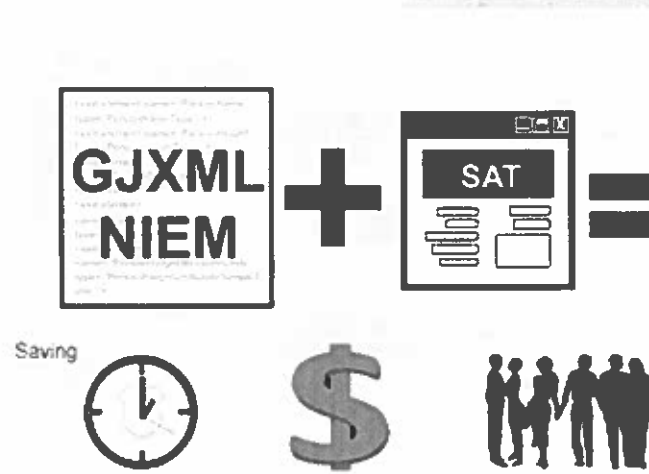
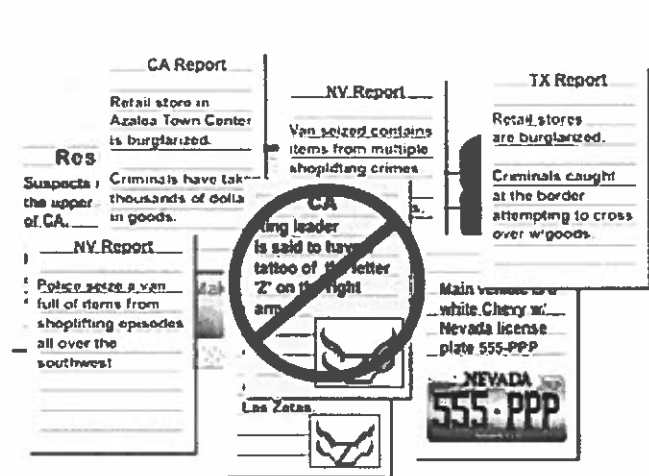
The video's total file length was a little over four and a half minutes. The video without audio was 172kb in size and with the audio it's 711kb in size.





Global Justice eXtensible Mark-up Language National Information Exchange Model Self-Assessment Tool

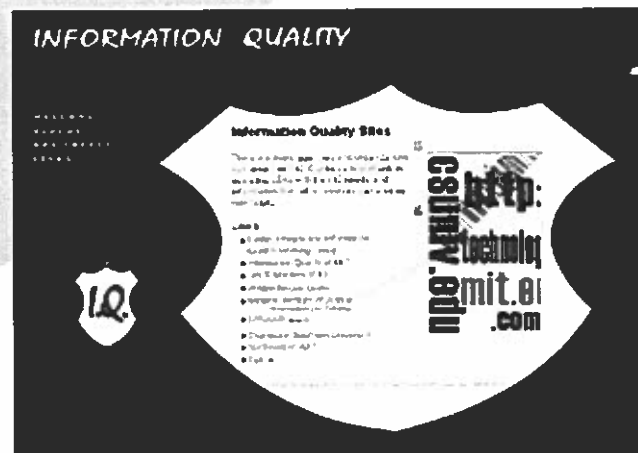
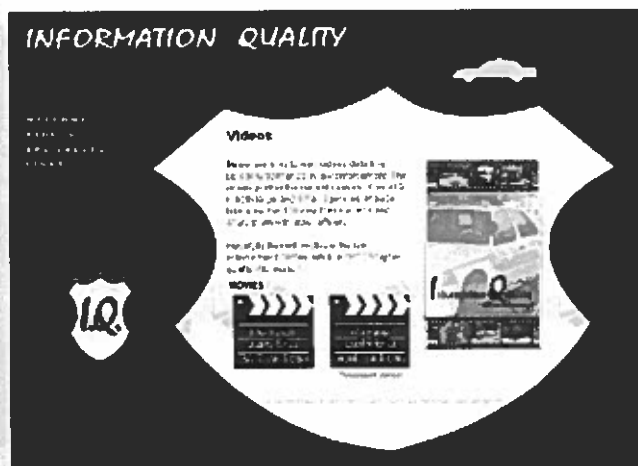
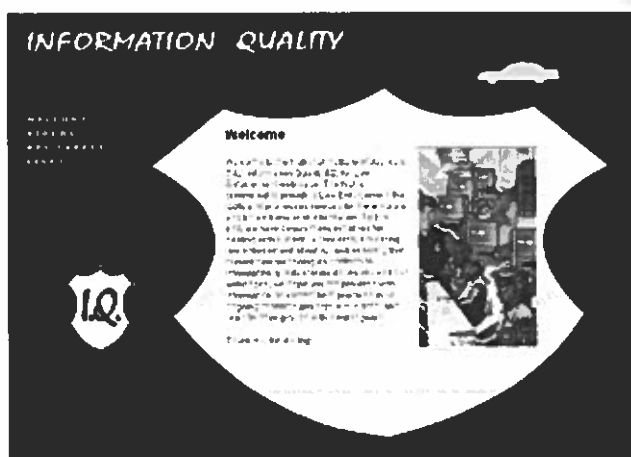




Part Three: Website

A website was created to hold the Information Quality videos created and the Best Practice Slick Sheets that were created by the Criminal Justice Department of CSU. Since there was not going to be a whole lot of content to include in the website, we decided to go with a more animated website. Since Flash is also known for its interactive site, we chose to use it. It was hard to come up with a nice design so we 'surfed' the web for ideas. We came across FlashVillage.com which contains lot of flash templates. There was one that was free to use for commercial or private use as long as you were a member of Flash Village. So

with becoming a member, we had our layout for our website. The original look of the template was of a map so we had to make it have a look more suitable for Law Enforcement. The reason for choosing that particular template was because the pages were only loaded if the view selected the page. So there was no need to load the whole website with both videos and the Best Practice Slick Sheets. With some editing to the template, we had our website.



Appendix A.

Two of the three basic scenarios used to create the script for the vignettes:

DENVER (AP) -- It was a routine traffic stop with the driver expecting a warning. But what he got was handcuffs and jail time. In the end, it was all a mistake, but as 7NEWS Investigator John Ferrugia discovered, the incident raises serious questions about how Aurora police conduct routine investigations.

Jerome Powell works for a government contractor and has no criminal record. He said he has had a security clearance ever since he was in the Navy. So how could he be the subject of an Aurora warrant for two felonies -- crimes he never committed?

He claims its sloppy police work by a veteran detective and police supervisors who didn't do their jobs. "He asks me to get out of the car and I thought that was a little weird, you know. I have never been asked to get out of the car after a traffic stop," said Powell. In late July, Powell drove through a yellow light as it turned red. A Mountain View police officer followed him and a block later pulled him over in a parking lot. After checking Powell's license in his computer, the routine stop turned scary.

"So I get out and I put my hands behind my back and he puts the handcuffs on me and says I am being arrested for an outstanding warrant," said Powell.

"I am shocked. I have no clue what's going on. I am not aware of any warrants out for me," Powell said.

Powell was being arrested for an incident in August 2003. During that incident, a man

walked into Aaron's Rental in Aurora and signed for more than \$10,000 in computer gear, furniture, and other expensive items, giving the employees Powell's driver's license as his identification. He then had the merchandise delivered to his apartment building but he never paid for it and never returned it. Powell had lost his driver's license in July of 2003 -- weeks before the theft-- and had it replaced.

"When I realized I lost my driver's license, I went to the DMV, told them it was lost and got a new one. I filled out the paperwork and got a new driver's license," Powell said. The Aurora police detective, using the information from the old license, began investigating Powell and seemed to ignore overwhelming evidence that Powell's lost license had been used by someone else. "I don't know how the officer issued a warrant out for me because everything pointed to the person that stole the rental equipment," Powell said. In fact, the apartment where the stolen items were taken was rented by a career criminal. The manager of that apartment told police he had never heard of Powell. The references and phone numbers listed on the rental contract are the career criminal's relatives -- no one related to Powell. And the career criminal didn't nearly fit Powell's description. "I outweigh him by 50 pounds. He's 5 inches taller than me. He has numerous tattoos. I have no tattoos," Powell said. "(The detective) had all this information but never called me or contacted me. I never got a letter (and he) never came to my house, never tried but somehow I got issued an arrest warrant."

"I knew it couldn't be him. There's absolutely no way. My son is a source of pride for the whole family. He was always a straight-A student. He went to Purdue. He studied nuclear engineering and has a nuclear engineering degree," said Powell's mother Yvette Corella. She just happens to be a Denver police officer. She called the Aurora detective, to talk

officer-to-officer, and look for answers.

"My son is listed in the phonebook. He's very easy to find. He's the only one listed with his name," said Corella. "(I asked,) 'You have his address. Why didn't you try to call him or have him come in? With or without counsel and let him know about the charges? This would have been cleared up two years ago. And (the Aurora detective) said he wasn't going to litigate this on the phone. That was it."

Acting Aurora Police Chief Terry Jones admitted his detective seemed to be fixated on Powell even though the evidence pointed elsewhere.

"He didn't do his job as thoroughly as I wish he had," Jones said.

Jones said if the detective had made a simple phone call to Powell he probably would not have been arrested. Everything that happened was a result of the detective not doing his job correctly, Jones said. "We're going to initiate an internal investigation into this, the reactions of the detective as well as the supervisor to find out what happened or what didn't happen," Jones said. "He's humiliated by the whole experience and just so upset. And I was trying not to cry and not to be upset so he wouldn't get upset. But he was just heartbroken, that anybody could think that he was a criminal," said Corella.

"The fact that I didn't do it and I did everything right. And I got arrested for something I didn't do. I mean, I actually went to jail," said Powell.

"Before, I never thought about going to jail. Now, it's in the back of my mind every time I see a cop -- 'I hope he doesn't pull me over,'" he said.

The Arapahoe County district attorney has dropped all charges against Powell, but he and

his family are out several thousand dollars after having to post bond and get a lawyer to clear his name. The Aurora police chief said he agrees Powell should be reimbursed for the department's error but he said that's a legal matter to be determined with the city attorney.

WARREN (AP) — A city woman arrested by Howland Township police on a warrant last month — and kept in Trumbull County Jail four days in error — says she believes she's been the ultimate victim of identity theft.

Renee A. White, 35, of Parkman Road, says that when Howland Patrolman Richard Mahan pulled over the car she was riding in with Michael R. Thomas, 18, about 2:19 a.m. Oct. 12, Mahan asked her about three false names that came up on the computerized law enforcement identification in connection with her name.

White says she told the officer during the traffic stop that she's never used an alias.

Specifically, the officer asked her about the name Tammy L. Johnson, who was wanted on an arrest warrant out of Liberty Township on two counts of forgery in the use of stolen checks.

"I'm like, 'No, I'm Renee White, and I've always been Renee White,'" she said.

Mahan then pulled up a picture of White on the computer in the police car.

"I said, 'That's my picture and yes that's my Social [Security number], but I'm not Tammy Lynn Johnson.' He's like, 'Well, you're under arrest for fraud,'" she said.

"I said, 'What are you talking about?' because I've never been in any trouble for anything other than driving."

White contends she continued to argue her case at the Trumbull County Jail, hoping the mistake would be caught there.

It wasn't.

She didn't have an attorney, and White said she fruitlessly sought help from family members.

Four days later, she was taken across the street to the Trumbull County Courthouse to be arraigned by Judge Andrew Logan of Common Pleas Court.

She once again said she was not Tammy L. Johnson, and Chris Becker, an assistant county prosecutor, checked out her story.

He took a photo of White on his cell phone and sent it to Liberty Police Detective Toby Meloro, who had interviewed Johnson during the investigation of the falsified checks, asking whether the woman in the courthouse was the right woman.

Meloro said it wasn't, and Judge Logan ordered White released.

"We got her out of there as soon as we knew what happened," Becker said, adding, "It's a real sad case of how far identify theft can go."

Becker and other officials say they still don't know what caused White to be arrested by mistake and are checking into it.

White says she thinks the reason the alias Tammy L. Johnson came up on the computer in connection with her is that she lost a state identification card over the summer at a gas station in Niles. She thinks Johnson may have found it and used it to steal her identity.

Detective Meloro says he doesn't know whether that happened. The name Renee White never came up during his investigation of Johnson, he said. White says she is not aware that anything was stolen from her as a result of Johnson's activities.

Chief Deputy Ernie Cook of the Trumbull County Sheriff's Department says White's theory of how she was falsely imprisoned is plausible. He's aware of cases in which the victim of identity theft has been accused of another person's crimes. He doesn't know if that happened here.

The real Tammy L. Johnson, 39, of Madison Avenue in Liberty, was arraigned before Judge W. Wyatt McKay on Wednesday. She is free on \$5,000 bond. If convicted on the charges, she faces up to two years in prison.

As for jail officials not realizing that White was not Johnson, Cook said the primary responsibility for arresting the correct person lies with the police officer. "We're just the holding facility," Cook said.

Dan Lester, assistant warden at the jail, said jailers working Oct. 12 accepted the explanation Mahan gave that White was the right person because of the alias.

If the police officer tells jail officials they are uncertain of a person's identity, then jail officials can check further — such as using fingerprints, Lester said. In this case, there is

no indication that Mahan or White raised enough suspicion to check further, Lester said.

Mahan could not be reached to comment. But Frank Dillon, assistant Howland police chief, said it appeared Mahan had "bent over backward" to make sure he had the right person by comparing her with a photograph on his computer.

The report from White's arrest says Mahan identified White as "Renee A. White, and it [computerized identification] showed that there was an active warrant for her arrest." There was no mention in the report of White's being arrested because she was identified as having an alias.

"Our officer acted on the information provided," Dillon said. "Everybody we arrest says, 'It's not me.'"

Appendix B. SOW.

STATEMENT OF WORK

FOR CHARLESTON SOUTHERN UNIVERSITY

IN SUPPORT OF THE

INFORMATION QUALITY Assessment ToolKit

BOM#

PERIOD OF PERFORMANCE

START DATE: Upon contract award

COMPLETION DATE: August 15, 2008

SHORT TITLE: INFORMATION QUALITY STUDY

1.0 PLACES OF PERFORMANCE

- a. Charleston Southern University, Charleston, SC
- b. Multiple Law Enforcement Agencies in South Carolina, Florida, Virginia
- c. Space and Naval Warfare Systems Center (SPAWARSYSCEN), Atlantic, Charleston, SC
- d. National Institute of Justice Headquarters, Washington, DC

2.0 REFERENCES: N/A

3.0 SPECIFICATIONS: N/A

4.0 SECURITY REQUIREMENTS

Contractor's request for visit authorization shall be submitted in accordance with DOD 5220.22M (Industrial Security Manual for Safeguarding Classified Information) not later than one week (5 business days) prior to visit. Requests shall be forwarded via SPAWARSYSCEN, Attn: Security Office, for certification of need to know by the specified COR/DOCOR. DD-254 of the basic contract applies.

It is not expected that any information under this study shall be classified higher than UNCLASSIFIED.

5.0 COR DESIGNATION

COR for this delivery order is Mr. Richard Baker, 843-218-4437.

ACOR for this delivery order is Mr. Dominic Lamar 843-218-4726.

6.0 DESCRIPTION OF WORK

6.1 SCOPE

SPAWAR, Atlantic is tasked with the development of an information quality toolkit to serve as a guide for law enforcement agencies that are striving to improve the quality of their data. SPAWAR, Atlantic, with help from its university partners, recently performed an information quality (IQ) study related to Sex Offender data sets. The purpose of this study was to develop a methodology for future IQ studies and to develop a baseline for IQ improvement research. Using a method adopted from the Department of Defense (DoD), Total Quality Data Management (TDQM), there are four steps to an IQ project – *define, measure, analyze, and improve*. Phase I research focused on the definition of data quality metrics and requirements for Law Enforcement and an initial measurement or IQ assessment of the current state of SC's Sex Offender data. Through this process, SPAWAR developed a methodology for assessing LE data that is transferable to other LE data sets.

Charleston Southern University (CSU) proposes to support and enhance this effort by performing the following tasks:

1. Collection and Documentation of IQ Best Practices

- a. SPAWAR, Atlantic shall work with the National Institute of Justice to determine LE agencies with outstanding practices in IQ. These may include the Area Regional Justice Information System (ARJIS) in the San Diego area, Florida Law Enforcement (FDLE), the Commonwealth of Virginia, and various local agencies in South Carolina and throughout the nation.
- b. CSU will then work to capture these best practices and develop one-page "slick sheets" that highlight these practices for distribution to the LE community at large.

This information will be developed with the intent to be distributed to the LE community; therefore they shall be succinct and adaptable to online presentation via the NIJ website or other suitable site.

- c. CSU shall develop this material and shall work with SPAWAR to determine the best distribution method for this material.
- d. CSU shall prepare a final document capturing all of the best practices, the contributing agencies, and a more in-depth look into the agencies' methodologies. This report will be suitable for research use by the Justice community as well as the IQ community.

2. Realistic IQ/ Decision Making Scenario

- a. CSU shall create a LE scenario that highlights the many nuances of poor IQ and its effect upon the decision making process.
- b. This scenario shall involve many aspects of IQ, most notably problems that can occur when high quality data is paired with low quality data, the effects of low quality data on LE decision making tools, and how reputation affects the usage of data sets.
- c. Every effort shall be made to incorporate and reference the standards found within the Law Enforcement Information Technology Standards Councils' (LEITSC) functional specifications for Records Management Systems (RMS) document, as well as the advances made by the Global Working Group on Privacy and Information Quality (GWGPIQ).
- d. The initial scenario shall be approved by the NIJ and then implemented by partners at Charleston Southern University.
- e. The final scenario shall involve an interactive media presentation created for presentation to the LE community during conferences. It shall also be developed with the intent to be shared via the web with the LE community at large.
- f. The NIJ and SPAWAR shall oversee the final presentation and scenario and shall determine its most suitable distribution method.

6.2 REQUIREMENTS

Provide the law enforcement/criminal justice research community with practical information on potential improvements to the IQ of LE data sets.

6.3 TASKS

1. Through field interviews utilizing both Criminal Justice and Computer Science research assistants, CSU shall capture LE best practices in IQ.
2. CSU shall develop material, suitable for distribution at conferences and via the Internet, that summarizes and highlights these practices. These shall be suitable for a LE audience.
3. Using free or low cost decision making software, CSU shall develop and implement a LE scenario that showcases the effects of poor IQ on decision making tools. Every effort shall be made to incorporate and reference the standards found within the Law Enforcement Information Technology Standards Councils' (LEITSC) functional specifications for Records Management Systems (RMS) document, as well as the advances made by the Global Working Group on Privacy and Information Quality (GWGPIQ).
4. CSU shall develop a multimedia presentation that walks through the scenario with the intent of convincing LE that IQ is an integral part of the decision making process.

7.0 GOVERNMENT FURNISHED INFORMATION: N/A

8.0 GOVERNMENT FURNISHED MATERIAL: N/A

9.0 GOVERNMENT FURNISHED EQUIPMENT: N/A

10.0 CONTRATOR FURNISHED EQUIPMENT: N/A

11.0 CONTRACTOR FURNISHED MATERIAL:

The contractor shall provide all support hardware, software, and expertise for the completion of this tasking.

12.0 TRAVEL REQUIREMENTS: Travel to/between Places of Performance authorized.

13.0 TRANSPORTATION OF EQUIPMENT/MATERIAL: N/A

14.0 DATA DELIVERABLES

Items produced under this SOW shall be forwarded to:

Space and Naval Warfare Systems Center, Charleston

Attn: Mr. Dominic Lamar(Code 56410DL)

PO Box 190022

North Charleston, SC 29419

14.1 Deliverable 1: Field Interviews

Meeting Notes for each Field Interview conducted during this study shall be included in the Monthly Reports. These meeting notes shall include the interviewees name and position, date, time, and duration of meeting, as well as a short synopsis of the conversation.

14.2 Deliverable 2: "Slick Sheets" Highlighting LE IQ Best Practices

"Slick sheets", or a SPAWAR approved presentation alternative, shall be developed that highlight best practices in LE. Final approved sheets shall be delivered to SPAWAR in the format of a Microsoft Word or Adobe Acrobat PDF document.

14.3 Deliverable 3: LE IQ Scenario

The LE IQ scenario shall be delivered to the SPAWAR representative. All software, data files, and documentation that is required to use the scenario shall be delivered in its native format.

14.4 Deliverable 4: IQ Scenario Presentation

The IQ scenario shall be developed with the help of the SPAWAR representative and a soft copy shall be delivered to the SPAWAR representative. This shall be delivered in the format of a Microsoft Powerpoint Presentation. Any accompanying files, software that are needed are to be included with this deliverable.

14.5 Deliverable 5: Administrative Reports

Monthly reports: Monthly reports for the previous month shall be submitted by the 5th day of each month in the format of Microsoft Word or Adobe Acrobat PDF documents, consolidating and detailing all tasking and expenditures for the month.

14.6 Deliverable 6: Final Study Report

A study report shall be created documenting the findings of the best practices study and IQ scenario. This final study shall be more detailed than the slick sheets and developed for an audience of researchers in the criminal justice or IQ field. The format of this report shall be approved by the SPAWAR representative and will be suitable for a publication in a practitioner's journal.

15.0 SUBCONTRACTING REQUIREMENTS: N/A

16.0 ACCEPTANCE PLAN

Acceptance of this tasking shall be contingent on approval from SPAWAR of the data deliverables detailed in section 14.

17.0 OTHER CONDITIONS/REQUIREMENTS: N/A

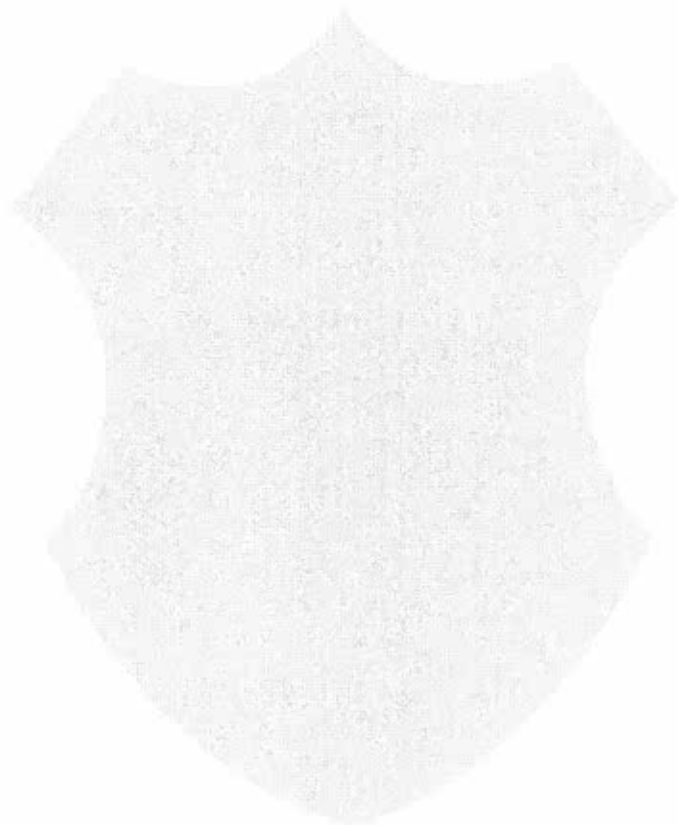
18.0 LIST OF ATTACHMENTS: N/A

Appendix C.

Final Report.

PDF Link:

[Final Report CSU ILP DQ Management.pdf](#)



Information Led Policing Data Quality Management

Final Research Report

**Prepared for the Space and Naval Warfare
Systems Center (SPAWARSYSCEN), Atlantic**

**Attn: Richard Baker
SPAWARSYSCEN, Atlantic
P.O. Box 190022
North Charleston, SC 29419**

**Charleston Southern University
Departments of
Computer Science and Criminal Justice
Charleston, South Carolina 29423**

Information Led Policing Data Quality Management

by Valerie Sessions and Jacqueline Fish

August 2008

Valerie K. Sessions, Ph.D., is an Assistant Professor in the Computer and Information Sciences Department at Charleston Southern University. Jacqueline T. Fish, Ed.D. is Chair of the Criminal Justice Department and Director of Criminal Justice Graduate Studies at Charleston Southern University. The research conducted by Charleston Southern University, Charleston, South Carolina, was supported by contract number N65236-08-P_6228 awarded by the Space and Naval Warfare (SPAWAR) Systems Center, Atlantic. The findings, conclusions and opinions expressed in this document are those of the authors and do not necessarily reflect the official position or policies of the National Institute of Justice, SPAWAR, Atlantic or Charleston Southern University.

Acknowledgements

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Dr. James Colman, Charleston Southern University
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Chapter 1 Introduction

“Computers are excellent tools – when they work” this phrase is heard daily no matter what the venue or profession of the speaker. Technology permeates our existence and every system we utilize relies on data integrity. Law enforcement agencies also rely on technology and when those tools are used incorrectly or become outdated, the effectiveness and efficiency of service to the Public also fails. Chief Executive Officers recognize not only the importance of investing financial resources into the systems that operate but just how critical quality data is to their mission to “serve and protect” our nation’s communities. The data that feeds these systems must be high quality, consistently accurate, and always improving. While leaders implement Total Data Quality Measures (TDQM) into their agencies, data integrity must be a priority for accurate analysis and utilization of technology in today’s ever-changing environment.

The Importance of Data Integrity

In 2000, the Bureau of Justice Statistics census identified 17,784 full-time law enforcement agencies (LEA) which provide more than one million full time employees who may be entering data into law enforcement databases. The sheer volume of data that is generated on a daily basis even at small agencies can quickly overwhelm and defeat records management systems (RMS) that are not designed with internal data integrity checks. The expense and lengthy process of selecting and implementing an RMS requires an enormous commitment on the part of the Chief or Sheriff, the funding governmental entity, and the employees (sworn officers or civilians) who must enter the data that keeps our criminal justice system operational.

Basis for this Study

The Total Data Quality Management process (TDQM) developed by the Department of Defense and Massachusetts Institute of Technology bases its success on user buy-in. Previous studies indicated that Chief Executive Officers (CEO) and records division supervisors (RDS) were highly supportive and understood the importance of data integrity. However, the need existed to identify best practices established by LEAs in an effort to increase the integrity of their data in every aspect of the operation of the central record collection operation. The data gathered by LEAs is the beginning point of all criminal investigations and this information is shared across the entire criminal justice system (CJS). The successes and pitfalls experienced by LEAs, regardless of the number of officers or the dollar value invested in systems must be made available to all other LEAs if we are to systematically increase data quality, and therefore enhance the ability of agencies to better serve their communities. This ability to share information via the Internet and the production of written materials is essential to increasing the accuracy of crime data as CEOs are faced with the need to increase the productivity of limited human and financial resources while the crime rates remain “the” measure of agency success. It has been recognized in many studies that LEA buy-in is one of the biggest attributors to the success of a data quality initiative. Therefore as an addition to the Best Practice study,

a set of web videos and instructional web pages was developed to assist the National Institute of Justice in educating law enforcement as to the far reaching effects of data quality problems. Two videos were envisioned – a first that targets large, technologically-heavy agencies in large urban areas, and a second that highlights the nuances of data quality problems in small agencies. It is the hope of the authors that these videos will inform agencies of the far reaching effects of data quality errors and provide an impetus for moving forward with data quality measures. Screen shots of the developed web pages and large agency video are included as Appendices F, and G of this report and were made available as a live test web page to our SPAWAR, Atlantic sponsors.

Study Oversight

Criminal Justice Department at Charleston Southern University

The principle investigator for the Information Led Policing Data Quality Management study (DQM) has the unique qualification of having held a position as a criminal records unit supervisor as a sworn law enforcement officer as well as working for a systems integrator to assist law enforcement agencies as they developed proposals for purchasing records management systems. Currently, as a department chair for an institution of higher education, Dr. Fish is able to blend her expertise from the researcher aspect with those of a law enforcement professional as well as a experience as a records management/integration specialist vendor. These credentials helped to establish credibility with the participating agencies, and generated a rich dialogue of interview data. The study results have yielded insight into critical decisions that must be made at the executive and operational levels if LEAs are to increase the quality of the data on which they rely to provide law enforcement services.

Research Assistants

Students majoring in criminal justice at Charleston Southern University were invited to submit applications for the two research assistant positions and after an interview process, review of transcript, and faculty recommendations, two outstanding candidates were selected for this project. The students began their research by studying the principles of Total Quality Management and law enforcement agency operation. To enhance their understanding of the processes that would be included in this study, they participated in agency ride-alongs and organizational tours. They submitted sample questions for the interview instrument, and observed the first agency interview to become comfortable with their responsibilities of conducting telephone interviews of participant agencies. The research assistants also compiled the lists of potential agencies and presented their data to the principle investigator for review. Twice weekly meetings were conducted and the list of potential agency participants was completed.

Computer and Information Sciences Department at Charleston Southern University

The Computer and Information Science (COINs) department at Charleston Southern is experienced in data quality assessments and research projects in this field. The primary researcher, Dr. Valerie Sessions, has conducted previous research in data quality in sex

offender studies, as well as incorporating data quality assessments into machine learning algorithms. She has been invited to speak on data quality at many NIJ and IACP conferences. Previously, she was also involved in data sharing and warehousing projects with the Naval Criminal Investigative Service (NCIS) and Department of Homeland Security (DHS). This experience has guided her efforts in law enforcement research and in creating the educational web pages and videos for LEAs.

Research Assistants

Two senior research assistants were chosen to create the web pages and videos. These students were experienced in developing multimedia web pages using technologies such as Flash™. The students also worked closely with the Information Technology professionals at Charleston Southern to create a professional web page design. The students also were able to gain insight from a private video services firm in Charleston that helped with the production of one of the videos. These assistants met as needed and were guided by the COIN primary investigator.

Chapter 2: Methodology for this Study

The quality of any study and the meaningfulness of the results rest on the design and implementation of the data collection tool combined with the expertise of the data analyst. This research was conducted utilizing a mixed methods approach. Quantitative data was collected through the use of a survey instrument, and qualitative data was generated by conducting personal interviews with the RDS designated as the point of contact by the chief executive officer.

Participant Selection

Initially, the study design included only law enforcement agencies that had attained the recognition of being awarded accreditation by the Commission on Accreditation for Law Enforcement Agencies (CALEA). Utilizing the CALEA website, agencies representing a geographic "spread" nationwide as well as the size of the agency were contacted via written communication and requested to participate in the study. A very low positive response rate dictated that we eliminate the accreditation requirement, and an additional group of agencies was selected based on location, size, and type of agency.

A total of 108 law enforcement agencies were invited to participate in this study with a response rate of 30% or 32 agencies. Police departments comprise the majority of LEAs nationwide, and they equal 72% of total respondents (city and metropolitan agencies). Sheriff's offices, as well as state universities and law enforcement organizations were among the agency participants.

The transferability of the results of this survey is expected to be very high due to the variety of responding agencies and the manner in which the interviews were conducted. Research assistants were cognizant of the agency characteristics through a process of familiarization including a study of websites and discussions with the principle investigator prior to conducting the interviews.

Respondent Demographics

Of the thirty two agency participants 23 were local/metro police departments which comprise the majority of law enforcement agencies. Three respondents were Sheriff's offices, and three were state university police departments and three were state law enforcement agencies. The geographic distribution consisted of three agencies in the Northeast (Delaware, New Hampshire and Pennsylvania), seven in the South (Alabama, Florida, South Carolina, Kentucky and Tennessee), nine agencies in the Midwest (Arkansas, Illinois, Indiana, Kansas, Michigan, Missouri, Ohio), six in the Southwest (Arizona, Texas, Colorado) four in the West (California, Hawaii, and South Dakota), and one in the Northwest (Oregon).

Agency size is defined as the number of sworn officers per agency. Respondents in this survey ranged from those classified as large (more than 100 sworn officers) to small

university police departments. A spreadsheet of participating agencies is provided in Appendix A for reference purposes.

Personal interviews were conducted with the supervisor or manager of the central records unit in each agency. These personnel were designated as a point of contact for this study by the chief executive officer of the agency.

The Survey Instrument

The quantitative data collection was conducted by using a survey instrument developed by the principle investigator and based on research conducted by Criminal Justice students. The initial instrument was reviewed by professional colleagues who are full-time sworn police officers and then was used on the first field interview led by Dr. Fish with the Summerville South Carolina Police Department. After completing this first interview, the questions on the survey were validated as capable of eliciting the proper type of response from our study participants. The survey instrument is available in Appendix B.

Initial Agency Contact

An introductory letter to the chief executive officer of 51 agencies was mailed out in April 2008. Enclosed with this letter were the following items: a printout of an article published on the Charleston Southern University website to establish credibility of the study, the survey instrument, a faxback letter, and a business card for the principle investigator. Copies of these documents are available in Appendix C of this report. The faxback form was designed as a simple means for the agency head to designate a point of contact and signify his/her agreement to participate in the study.

After two weeks, agencies that had not responded to the initial invitation to participate were mailed a reminder postcard in an attempt to generate more respondents. Lacking a sufficient response rate from CALEA accredited agencies, we widened the scope of our potential study participants while attempting to maintain an equitable geographic distribution. The initial invitation package was sent to 57 additional agencies. The same two week reminder postcard was sent to non respondents and the result was 32 of 108 agencies contacted agreed to participate in the study.

Subsequent Steps

Upon receipt of the faxback form, Research Assistants were assigned to complete the interviews. The point of contact (POC) for each respondent was contacted and a convenient time was scheduled for the interview to take place. In a few instances, the responding agency e-mailed the answers ahead of time so the interviewers could be more focused on specific information on which they wanted to expand their inquiries. This is one of the benefits of qualitative research—the opportunity to establish a rapport with the respondent provides enhanced data that in turn, leads to a greater understanding of the problems and successes of their data quality programs.

The research assistants and the principle investigator conducted 32 interviews during June and July 2008. Weekly meetings among the researchers provided a time to review the interview results and determine the need for follow-up telephone calls or e-mails. All data was submitted to the principle researcher for analysis and development of the final report.

Chapter 3: Analysis of the Data

The overall target population of 108 agencies was selected to properly represent the relevant subgroups of LEAs. These subgroups included police and sheriffs agencies, local and state funded agencies including university police departments, and an equitable geographic distribution. Those agencies that chose to participate did provide us with a cross-sectional study that reflects both the positive and negative experiences of RMS personnel who occupy the position of senior management responsible for the agency's data collection.

Quantitative Results

Mobile Computing

The most prevalent technological advance among respondents is the use of laptop computers by street officers. Eighty-eight percent of the agencies report their officers utilized mobile data terminals (MDT) in their police vehicles. The uses of laptops or MDTs varied considerably with most being used for communications regarding vehicle registrations, driver license checks, warrants or stolen property (NCIC) or telecommunications from central dispatch. Of that 88%, more than half also use the laptops or MDTs for generating online offense, incident, or accident reports. One problem encountered by agencies that were seeking to computerize all reports was the requirement that in some states certain types of reports (missing persons, fraud and identity theft) require original signatures of the complainant therefore dictating the need for handwritten reports created by the responding officer and signed by the complainant at the time the report is initiated. Other handwritten reports are created by private prosecutors such as loss prevention personnel working with retail stores to curtail shoplifting and employee theft. While police officers transport these arrestees to booking facilities, the store employees must generate the arrest and offense reports that are then taken to the booking facility with the prisoner.

It is also interesting to note that even motorcycle officers are using portable technology to generate reports. Many agencies reported their officers use smaller laptops or other mobile devices to collect data and create official reports.

Online Report Generation

Slightly more than half of the respondents (56%) require their officers to file their reports electronically. While all these officers do not have laptops in their vehicles or they are not using those MDTs for reporting duties, the agencies have desktops or laptops available in precinct locations for the officers to access in order to complete their reports. The reports are then uploaded via satellite or they are transferred on a flash drive to the central records unit for processing. It is equally reported that 18 agencies still use handwritten reports in some or all offense, incident, or accident reporting.

Current Records Management Systems

Two thirds of the respondents' records management systems were more than 5 years old. Ten agencies reported their systems were new or it had been less than 4 years since the systems were implemented.

The implementation time frame is defined as an estimate of the amount of time that passed from project inception to "going live" with the new records management system. We attempted to collect data in this category; however we encountered the problem of various supervisors in charge during different phases, and many respondents only 13 agencies (42%) could accurately respond to this question. Three agencies reported there was a 3 year or more time period while 7 agencies worked 18 months to implement a new system.

Entering Handwritten Reports into Records Management System

Of the agencies that still generate some form of handwritten report, 87% responded that data is entered into the records management system by personnel assigned to the records division. The majority of these personnel are civilian records clerks, however in some agencies, officers who are working "light-duty" assignments may also be entering this data.

Report Review Process

Whether reports are generated online in the field or handwritten, all are subjected to a series of reviews prior to the data being entered into the records management system. Online report generation significantly enhances the quality of the data as officers make pre-determined selections from drop-down menus or pre-defined fields. Built-in data integrity checks increase the accuracy of this data from the point of entry. Online reports are routed to the electronic in-box of the supervisor on duty for review who check for proper offense classification, that elements of the criminal offense are reported, and overall accuracy of the report. Once they have reviewed and approved the online report, it is released for entry into the database of the records management system. When errors are detected by the Supervisor, the report is re-routed to the initial officer for correction prior to being submitted to the database.

Handwritten reports are submitted to the Supervisor for review prior to being routed to the data entry clerks in the records division. The review process remains the same as the online report; if the reviewer detects errors, the report is returned to the initial reporting officer for correction prior to being submitted to the data entry clerks.

Data Quality Control

Automated process checks were cited as the most effective means of auditing data for accuracy. Many agencies reported including this feature in their standard software requirements; however several of the older systems do not have this feature and must rely on random checks conducted by the RDS. UCR/NIBRS reports systematically reject non

compliant coding, however the possibility exists that the offense was improperly classified by the reporting officer, misconstrued by the reviewing officer, and can skew the crime rates when undetected. Supervisors report that unexplained spikes in monthly reports generate a second review of the data to determine if there has been an error in coding, but this issue can go undetected for longer periods of time in agencies where "spikes" are not met with increased attention by crime analysis or other specialized units.

Other methods of checking for data integrity that were reported by the respondents included accreditation audit checks, the existence and extensive use of a Quality Control Unit, and random checks by records department personnel other than the supervisors.

Qualitative Results

The standard questions asked of all respondents yielded descriptions that validated the quantitative research and provided us with results that can be both tabulated and explained. Agency business practices, the operating environment, the financial status of the funding entity, and the compelling need to improve the quality of data being collected permeated these interviews. Important best practices to implement as well as faulty efforts to avoid were identified in this portion of the study.

Overwhelmingly, the consensus of the respondents is an increased need for training for all agency personnel. RDS echo the need for ongoing training efforts – the reporting requirements continue to evolve, technological advances make it possible to increase the quality and quantity of data in an efficient manner, and the integration of Geographical Information Systems (GIS) with Computer Aided Dispatch (CAD) and Records Management Systems (RMS) can enhance the effectiveness and efficiency of law enforcement agencies while contributing to an enhanced level of safety for police officers.

CEOs recognize not only the importance of investing financial resources in the systems that operate their agencies, but that the data that populates these systems must be high quality, consistently accurate, and always improving. When these leaders integrate TDQM principles into their agencies, organizational function is enhanced.

Specifically, the qualitative interviews revealed consensus among the agencies regarding best practices – or advice they wanted to share with other organizations in the process of planning to implement a records management system that focused on data quality. These findings were consistently repeated by agency representatives – whether large or small, urban or rural, and independent of geographic location.

- Technology is a wonderful tool as long as the agency uses the tool to solve problems rather than create new problems.
- Criminals do not follow jurisdictional lines – be sure your new system can share information with other agencies.

- Pre-define all the data fields in every module and ensure there is a single point of entry for this data.
- Never assume the data is what you thought it would be. Run random queries on the system to identify inconsistencies. Build in safeguards to avoid fictitious reports from being entered into the system.
- Identify and implement internal and external quality checks for data entry and integrity.
- Formulate a strategy to proactively analyze, plan, and implement technology – do not fall “behind the curve”.
- The elimination of redundant data entry, reduction of human intervention, eradication of paper, and access to multiple databases will greatly enhance the integrity of data and facilitate the retrieval of that information.
- Ongoing training keeps officers and investigators informed and enhances the probability the data collected is accurate and crimes are properly classified.
- Agency leaders must accurately assess the costs associated with maintenance of legacy systems against the investment in state-of-the-art database management systems.
- Business processes must evolve to accommodate enhanced technology, for example, accepting electronic signatures on documents instead of originals and eliminating the need for paper-based reports.
- Send records unit supervisors and daily users to observe comparable size “live” systems in operation that are utilizing the same modules as your agency will need. Do not make a purchasing decision based on a vendor’s demonstration of “canned” data.
- Obtain the source code from the vendor so that your internal computer systems managers can design additional queries as new demands for data analysis emerge.
- Don’t waste time and resources maintaining multiple systems. Ensure a single point of control and that data is shared among databases. Consolidate duplicate data when it is detected.

Chapter 4 Summary of the Study

Law enforcement executives are making the best use of scarce financial resources by integrating many forms of technology to supplement the lack of sufficient personnel. Data management and systems integration have proven to be effective tools that maximize law enforcement efforts to identify, apprehend, and prosecute offenders. The quality of the data that is gathered from the initial point of contact throughout the investigative process—and indeed even to the adjudication of the case and the correctional system is integral to the success of the Criminal Justice System.

One of the most essential components that contribute to data quality is that chief executive officers understand just how critical it is to gather precise information that becomes quality data. This educational process undergirds the entire Criminal Justice System and cannot be overlooked or relegated as a low priority item.

Sworn and civilian personnel are responsible for gathering and entering data into the electronic infrastructure of America's justice system. The importance of this task is paramount when evaluating methods to improve the data collection that drives our law enforcement efforts.

While information gathering of quality data is the foundation of successful law enforcement operations, sharing that information through electronic interfaces across multiple jurisdictions is essential to the safety of our communities. These efforts will be thwarted if new records management systems are not designed to integrate via secure communications networks to GIS, CAD, and the Integrated Automated Fingerprint Identification System (IAFIS).

Records division supervisors contributed many insights regarding successful implementations focused on data integrity and the necessity of conforming to the Global Justice SML (GJXML) data model. The input gained through this study resulted in the identification of Best Practices that should be heeded by agencies that will be upgrading or replacing legacy information management systems. At the same time, researchers were able to compile a list of "Worst Practices" which contains a set of actions to avoid when managing the planning and implementation of a new records management system. The pitfalls that were identified by the study respondents are as essential as the Best Practices and have been provided in Appendix D of this document.

Whether the agencies were large or small, and regardless of the type of agency or geographic location, Best Practices for ensuring data quality are universally applicable. Criminal offenders do not restrict their activities to jurisdictional boundaries, and many crimes are solved based on data gathered and shared at the local level. Automation of law enforcement data is as important to an agency with five officers as it is to an agency of five thousand officers.

First and foremost in the conversations with records division supervisors was the safety of police officers on the street. Data that is gathered in a haphazard fashion or data that cannot be retrieved because of original inaccuracies could place officers and citizens in potentially life-threatening circumstances. While fortunately these incidents are few, some agencies may not focus resources or emphasize the importance of data integrity. The value of credible data must be accentuated from the onset of every technology initiative. Reduction of crime while improving the quality of life in America's communities are the goals of local agencies and this is one huge part of the never-ending cycle that can be improved upon through the adoption of the Best Practices recommended by the respondent agencies in this study.

Limitations

Despite our best efforts, the response rate of agencies to the invitation to participate in this study did not meet our initial goal. The design of the study incorporated the various types of agencies as well as provided an equitable geographical and agency size distribution. Also, we had originally identified CALEA accredited agencies as the only source of participants; however the low response to our initial round of invitations forced us to drop this parameter. Because of the small response to the survey (30%) the results should not be considered scientific. However, the comments that led to the development of the Best Practices bullets in the previous section did self-validate as they were echoed by survey respondents multiple times throughout the interviews. These common threads linked agency representatives and their experiences and served as the basis for the Best Practices.

Future Implications

Technological advances continue to impact law enforcement from all aspects. Data integrity and information-led policing are essential components of the Criminal Justice System and it is critical that emphasis continue to be focused on practices that will enhance law enforcement's ability to respond to our calls for service. The continuity of responses gathered from the 32 participating agencies underscores their appreciation for data quality and their willingness to share both the best and the worst of their system implementations serves as an indicator of just how vital accurate information can be to the lives of our police officers, the effectiveness and efficiency of local agencies, and the commitment to clearing the innocent and convicting the guilty.

Information Led Policing Data Quality Management

Appendix

Appendix A Agency Spreadsheets

Information Led Policing Participating Agencies

Agency	City	State	Contact	Phone
Huntsville PD	Huntsville	AL	Capt. Vince Dauro	256.427.5652
Northern AZ Univ.	Flagstaff	AZ	D'Ann Reeths	928.523.1696
Univ of AR PD	Fayetteville	AR	Gary Crain	479.575.2311
Little Rock PD	Little Rock	AR	Mary Cook	501.371.4824
Riverside PD	Riverside	CA	Roz Vinson	952.353.7605
Sacramento PD	Sacramento	CA	Claudia Evans	916.808.0624
Colorado Springs PD	Colorado Springs	CO	Robert Kean	719.444.7646
Loveland PD	Loveland	CO	Teri Kuhlman	970.962.2265
Dover PD	Dover	DE	Patricia Virdin	302.736.7108
Jacksonville SO	Jacksonville	FL	David W. Vanderhoff	904.630.2135
Miami-Dade Co PD	Doral	FL	Heather Thomas	305.471.1863
Honolulu PD	Honolulu	HI	Aaron Correia	808.529.3111
Bloomington PD	Bloomington	IL	Jack McQueen	309.434.2228
Brownsburg Metro PD	Brownsburg	IN	Kimberly Shupert	317.852.1109x2114
Topeka PD	Topeka	KS	James Anderson	785.368.9444
Prairie Village PD	Prairie Village	KS	Tim Kobe	913.385.4620
Lexington Metro PD	Lexington	KY	James Decker	859.258.3671
Detroit PD	Detroit	MI	David Jones	313.475.1449
Michigan State Univ.	East Lansing	MI	Florene Mclothian-Taylor	517.353.8502
Missouri St Highway Patrol	Jefferson City	MO	T.G. Boysinger	573.526.6250
Lincoln PD	Lincoln	NH	Clint Smith	603.745.9000
Akron PD	Akron	OH	Mike Prebonick	330.375.2488
Portland Police Bureau	Portland	OR	Kimberly Roark	503.263.7666
Philadelphia PD	Philadelphia	PA	Thomas Olson	215.686.1479
Richland Co SO	Columbia	SC	Randy Scott	03.576.3033
SLED	Columbia	SC	Buddy Wilkes	803.896.6995
Summerville PD	Summerville	SC	James Bateman	843.871.2463
Rapid City PD	Rapid City	SD	Ray Cornford	605.355.3610
Knoxville PD	Knoxville	TN	Gus Paidousis	865.215.7229
Dallas PD	Dallas	TX	Scott Gerdes	214.671.3103
Plano PD	Plano	TX	Charlotte Holzmeister	972.941.2582
Texas Dept. Public Safety	Austin	TX	Todd Kresnik	512.424.2129

Appendix A Agency Spreadsheets (continued)

Invited Agencies

Agency	City	State
Birmingham PD	Birmingham	AL
Huntsville PD	Huntsville	AL
Jefferson Co SO	Birmingham	AL
Montgomery PD	Montgomery	AL
Anchorage PD	Anchorage	AK
Juneau PD	Juneau	AK
Northern AZ Univ.	Flagstaff	AZ
Phoenix PD	Phoenix	AZ
Univ of AR PD	Fayetteville	AR
Ft. Smith PD	Fort Smith	AR
Little Rock PD	Littlerock	AR
Beverly Hills PD	Beverly Hills	CA
Los Angeles PD	Los Angeles	CA
Modesto PD	Modesto	CA
Riverside PD	Riverside	CA
Sacramento PD	Sacramento	CA
San Diego PD	San Diego	CA
Colorado Springs PD	Colorado Springs	CO
Denver PD	Denver	CO
Loveland PD	Loveland	CO
Univ of CO	Boulder	CO
Berlin PD	Berlin	CT
Groton PD	Groton	CT
Dover PD	Dover	DE
FL Dept. of Law Enf.	Tallahassee	FL
Jacksonville SO	Jacksonville	FL
Miami-Dade Co PD	Doral	FL
Oak Hill PD	Oak Hill	FL
Atlanta PD	Atlanta	GA
Camden Co SO	Woodbine	GA
Decatur PD	Decatur	GA
Honolulu PD	Honolulu	HI
Maui County PD	Maui	HI
Pocatello PD	Pocatello	ID
Bloomington PD	Bloomington	IL
Eastern IL Univ	Charleston	IL
Cook Co SO	Maywood	IL
Illinois State Police	Springfield	IL
Brownsburg Metro PD	Brownsburg	IN
Indiana Univ.	Indianapolis	IN
Sioux City PD	Sioux City	IA
Topeka PD	Topeka	KS
Salina PD	Salina	KS

Invited Agencies (continued)

Agency	City	State
Prairie Village PD	Prairie Village	KS
Lexington Metro PD	Lexington	KY
Newport PD	Newport	KY
Louisiana State Police	Baton Rouge	LA
Danvers PD	Danvers	MA
Univ of MA	Amherst	MA
Auburn PD	Auburn	ME
Portland PD	Portland	ME
Rockville PD	Rockville	MD
Salisbury PD	Salisbury	MD
Detroit PD	Detroit	MI
Michigan State Univ.	East Lansing	MI
Anoka PD	Anoka	MN
St. Paul PD	St. Paul	MN
Jackson PD	Jackson	MS
MS State Univ PD	Mississippi State	MS
Missouri St Highway Patrol	Jefferson City	MO
Salisbury PD	Salisbury	NC
Lincoln PD	Lincoln	NE
Lincoln PD	Lincoln	NH
Lincoln Park PD	Lincoln Park	NJ
Oaklyn PD	Oaklyn	NJ
NJ State Police	West Trenton	NJ
San Juan Co SO	Aztec	NM
Albuquerque PD	Albuquerque	NM
Farmington PD	Farmington	NM
Conandrigua PD	Conandrigua	NY
New York PD	New York	NY
Rochester PD	Rochester	NY
Suffolk Co PD	Yaphank	NY
Charlotte PD	Charlotte	NC
Clayton PD	Clayton	NC
Conover PD	Conover	NC
Greenville PD	Greenville	NC
Bismark PD	Bismark	ND
Amerley Village PD	Cincinnati	OH
Akron PD	Akron	OH
Tulsa PD	Tulsa	OK
Grants Pass DPS	Grants Pass	OR
Portland Police Bureau	Portland	OR
Derry Township PD	Hershey	PA
Philadelphia PD	Philadelphia	PA
Bristol PD	Bristol	RI
RI State Police	North Scituate	RI

Appendix A Agency Spreadsheets (continued)

Invited Agencies (continued)		
Agency	City	State
Charleston PD	Charleston	SC
Greenville Co SO	Greenville	SC
North Charleston PD	N. Charleston	SC
Richland Co SO	Columbia	SC
SLED	Columbia	SC
Summerville PD	Summerville	SC
USC Div of Law Enf.	Columbia	SC
Lexington Co SO	Lexington	SC
Rapid City PD	Rapid City	SD
Collierville PD	Collierville	TN
East TN State Univ	Johnson City	TN
Knoxville PD	Knoxville	TN
Dallas PD	Dallas	TX
Plano PD	Plano	TX
Texas Dept. Public Safety	Austin	TX
Texas Southern Univ	Houston	TX
Virginia Tech PD	Blacksburg	VA
Oak Harbor PD	Oak Harbor	WA
WV State Police	South Charleston	WV

Appendix B Survey Instrument

Information-Led Policing Data Quality Management Survey Questions for National Institute of Justice Survey Charleston Southern University Criminal Justice Department June 2008

The focus of this project is to identify best practices developed and implemented by law enforcement agencies that have resulted in better quality information for your officers and investigators. These are the questions the researchers at Charleston Southern will be asking during your phone interview. Please feel free to elaborate on any aspect of the survey. Thank you in advance for participating in this NIJ study.

1. What innovations have your agency implemented that has increased your information gathering capacity?

Laptops in police cars

Online report generation

Handwritten

Other _____

2. How long has your current Records Management System been in operation? Is it fully functional or are you adding components on a scheduled implementation plan. What is the timeline for full implementation OR how long was your implementation program?
3. Since you have been utilizing your current RMS, what change of process has your agency implemented that has had the most positive impact on the quality of your data?
4. Are reports completed via computer in the field? Are the computers installed in police vehicles or do the officers utilize desktop units to complete and submit reports?
5. If handwritten reports are still being completed, who inputs the data into the records management system?
6. What is your process for report review and approval by patrol supervisors?
7. How does your agency perform quality control i.e. ensure the integrity of the data?
8. What is the single best piece of advice you can offer to other agencies who are striving to improve the quality of their information?
9. Is there anything you would do differently to improve the information gathering process?
10. If there was one thing you could do to change your current process to enhance the quality of data, what would it be?

Appendix C Contact Documents – Invitation to Participate

April 19, 2008

Chief Richard W. Meyers
Colorado Springs Police Department
705 S. Nevada Ave.
Colorado Springs, CO 80903

Dear Chief Meyers:

The National Institute of Justice (NIJ) has funded a nationwide project to identify *Best Practices in Information-Led Policing*. Your agency has been selected in a preliminary search of CALEA organizations as a potential participant in this study. There is no requirement for commitment of resources to be a participant agency. The study is designed to gather information through a short phone conversation (20 minutes) or face-to-face interview (30 minutes) with a Charleston Southern University Research Assistant.

The outcomes of the research will be published by the NIJ and featured on both informational “slick sheets” as well as an interactive website designed specifically for law enforcement organizations. Agencies whose Best Practices are selected for inclusion will be identified in the final report and the publications.

We are very excited at the prospect of speaking with your designated representative and have included a copy of the interview questions for your review. Your agreement to participate in this study requires only two action items on your part

1. Please designate an authorized spokesperson that can provide the data we are seeking regarding your records management and data collection procedures and forward these interview questions to your designee.
2. Complete and fax back the enclosed form with the contact information for your designee so that a Research Assistant can make arrangements to complete an interview.

We plan to provide an abstract of the research findings to all agencies if you indicate your interest in the outcomes on the enclosed faxback form. The study will be concluded by July 31 and the final products will be submitted to the NIJ in August 2008.

Your willingness to share your Best Practices with professional colleagues will assist in the enhancement of information and ultimately in more effective collection and dissemination of data and intelligence to law enforcement officers across the United States. We appreciate your cooperation and look forward to including your agency in our study. If you have any questions, please contact me at 843.863.7131 or via e-mail: jfish@csuniv.edu.

Sincerely,

Jacqueline T. Fish, Ed.D.
Chair, Criminal Justice Department



CHARLESTON
SOUTHERN
UNIVERSITY

Department of Criminal Justice
FAX-BACK FORM
843.863.7198

Agency: _____

I am pleased to accept the invitation for my agency to participate in the National Institute of Justice study on *Information-Led Policing* currently being conducted by the Criminal Justice Department of Charleston Southern University. I am providing the point of contact information below for the researchers so they will be able to conduct the survey with my appointee.

Point of Contact: _____ Phone: _____

E-mail address: _____

Agency Executive

Date

Charleston Southern University: Dr. Jacqueline Fish 843.863.7131

Criminal Justice Department awarded Grant - April 14, 2008

Added by: University Relations

Charleston Southern University has been awarded a grant by SPAWARSCEN, Charleston, S.C. to conduct research in the area of information quality related to law enforcement agencies. As technological advances enhance the ability of agencies to share information and build intelligence, the quality of that data is of paramount concern. Local and state agencies utilize shared information to guide decision making, positively identify people, and make appropriate arrests.



Pictured: (L to R) John Simkulet, Jasmin Robinson and Dr. Jacqueline Fish

“We are very fortunate to partner with SPAWARS,” said Dr. Jacqueline Fish, director of graduate studies and chair of the criminal justice department at Charleston Southern, “it is a wonderful opportunity for our students to gain first-hand experience working with local, state, and federal agencies while they are still pursuing their undergraduate degrees.”

The National Institute of Justice (NIJ) is the research, development, testing, and evaluation agency of the United States Department of Justice. They have awarded an Information-Led Policing grant to SPAWARDS to continue previous research conducted on the quality of information and data utilized by police agencies to build intelligence that will assist in the reduction and prevention of crime.

The Computer and Information Sciences (COIN) department, along with the Criminal Justice (CJ) department at Charleston Southern will complete a major part of the grant deliverables under the direction of Principle Investigator Dr. Jacqueline Fish. She will forward all findings and products to SPAWARS Program Manager Richard Baker for approval and submission to NIJ.

Charleston Southern student research assistants will gain valuable knowledge and experience gathering data and developing deliverables for the project. Senior Steve Garbeil, a software engineer from Summerville, will create a web-based scenario depicting the consequences and outcomes of a law enforcement scenario based on substandard data. Steve is working on this assignment as a part of his Senior Project.

Rebecca Randall, a senior from Moncks Corner, also a COIN student, will assist the project by designing a web page and helping with both the scenario development and best practices data development.

Juniors Josh Simkulet of Summerville and Jasmin Robinson of Summerville are Criminal Justice majors who will be gathering “best practice” data from law enforcement agencies and compiling the results of this results of this research into publicity “slick sheets” designed specifically for the law enforcement audience. “I am really enjoying the project

and the opportunity to get first hand accounts of how various agencies are operated,” said Robinson, “hopefully I can participate in another study in the future. I am also looking forward to the finished phase to see how everything I’ve done comes together and works out.”

“All of the Charleston Southern students are enthusiastic about working on a project that will be submitted to the NIJ,” said Dr. Fish, “the results will positively contribute to the improvement of data quality and the future actions of local and state enforcement agencies.”

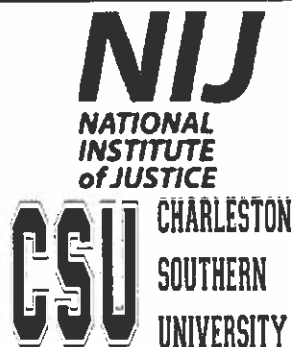
At the conclusion of the project, the hard work of the research assistants will be featured on an interactive website and in publications designed for law enforcement agencies across the United States.

INFORMATION LED POLICING PROJECT

National Institute of Justice and Charleston Southern University

Recently your agency was offered the opportunity to participate in a nationwide survey to determine best practices in information collection and dissemination. Please take a moment to complete and submit the faxback form so that researchers will be able to include your department in this important study.

We have selected outstanding law enforcement organizations and your expertise is needed to assist other agencies in developing higher quality data for investigative purposes! Phone survey—20 minutes—that is all it requires for your agency to be included. CALEA accreditation is not required for participation. **THANK YOU!**



Dr. Jacqueline Fish
Criminal Justice Department
Charleston Southern University
9200 University Blvd.
Charleston, SC 29461
Phone: 843.863.7131
Fax: 843.863.7198
E-mail: jfish@csuniv.edu

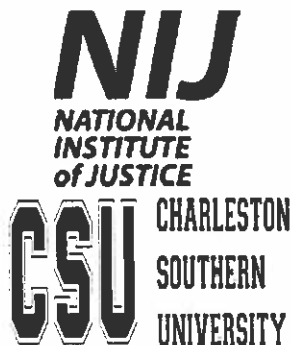
An invitation package was mailed to your agency on May 24, 2008.

INFORMATION LED POLICING PROJECT

National Institute of Justice and Charleston Southern University

Recently your agency was offered the opportunity to participate in a nationwide survey to determine best practices in information collection and dissemination. Please take a moment to complete and submit the faxback form so that researchers will be able to include your department in this important study.

We have selected outstanding law enforcement organizations and your expertise is needed to assist other agencies in developing higher quality data for investigative purposes! Phone survey—20 minutes—that is all it requires for your agency to be included. CALEA accreditation is not required for participation. **THANK YOU!**



Dr. Jacqueline Fish
Criminal Justice Department
Charleston Southern University
9200 University Blvd.
Charleston, SC 29461
Phone: 843.863.7131
Fax: 843.863.7198
E-mail: jfish@csuniv.edu

An invitation package was mailed to your agency on May 24, 2008.

Appendix D Worst Practices List

- ✓ Remember to identify all your users and supply printers where needed – don't forget the booking facility needs access to the information.
- ✓ Failure to identify all points of data entry where handwritten reports are still accepted can be embarrassing. This includes loss prevention arrests from retail establishments and documents requiring original signatures such as fraud statements.
- ✓ What is your fallback plan when there is a power failure and your online report generation system cannot transmit data?
- ✓ Build in safeguards that will eliminate phony reporting entries if you utilize an Internet based online citizen reporting module. Kids can have a field day with this one.
- ✓ Remember the information is only as good as the officer or data entry clerk types. Random spelling checks are very useful.
- ✓ Software and data element fields must be updated at the same time as the statutes change (not six months later because that's when it is scheduled).
- ✓ Don't let people with a vested interest in the crime rates categorize the crime data.
- ✓ It is very expensive and time consuming to maintain legacy systems with patches. This is like putting a band-aid on a failing bridge – you are going to get soaked by the vendor.
- ✓ If you only train your personnel one time – your data collection records will reflect this poor decision.
- ✓ Transitioning personnel through the Records Division in order to give them supervisory experience will not be beneficial to your agency. Strive for stability and build leadership in your records unit supervisors.
- ✓ More data is good; more technology is excellent; more errors will haunt your database for a long, long time.
- ✓ Don't sacrifice quality for quantity.
- ✓ Dispatchers need to be able to add information to the permanent files even after the officers have cleared the scene. Do not lock them out – this creates even more problems as they have to create supplementary information screens which may not be accurately linked to the original incident number.
- ✓ Do not let the Sheriff's brother-in-law supply your next records management system.
- ✓ Most importantly – do not allow officers to become so immersed in the technology available on their laptops that they let down their guard and get injured on the street.

INFORMATION LED POLICING

TECHNOLOGY

TOOLS

"Data entry and recording problems are common and well known, and, unfortunately, detectives and police officers often ignore the details of a file because they know that the data are unreliable."

Jerry Ratliff in
*Information Led
Policing*



Data Quality Best Practices

There's nothing worst than finding out at the end of a long crime spree that your agency had the knowledge and the power to save lives and property through early identification, but was unable to do so due to corrupt data. Inevitably criminals reside in our data long before they're in our custody. Maintaining a clean and usable database is just as important as having a functional firearm or police vehicle when they're needed most. I can't imagine why any agency would continue to collect data they couldn't use!

Major Ronald D. Hairston, City of Danville, VA Police Department
2009 IACP LEIM Section Chairman

Data Integrity Benefits...

- Maximize the effectiveness of manpower deployment and resource allocation
- Enhance police officer and community safety
- Increase officer awareness of crime pattern information
- Reduction of errors in data entry and retrieval
- Improve linkages across the criminal justice system
- Decrease law enforcement response time to calls for service
- Simplify record retrieval for all authorized users
- Use technology to solve problems rather than create new ones

Best Practice Slick Sheet Version I Front

Best Practices Identified by Agency Participants

- Select integrated CAD, RMS, GIS, and maintain ONE database
- Utilize multiple query fields to provide numerous search capabilities
- Pinpoint crime hot spots with enhanced query capabilities
- A Quality Control Unit is critical
- Evaluate live systems—do not rely on demo databases
- Pre-planning is essential; identify all stakeholders early in the process
- Obtain the source code for future upgrades
- Identify and implement internal and external quality checks for data entry and integrity
- Run random queries to identify inconsistencies
- Training—training—training is essential to keep the entire agency informed
- Eliminate paper in every process

Survey Comments

- * Online reporting = 100% legibility!
- * Implement advances in technology as they occur
- * The ability to browse and query various fields allows officers to identify patterns and link suspects to offenses
- * Criminals don't follow jurisdictional lines; quality information can be shared among agencies and provide leads to solving multiple crimes
- * Automated reporting reduces UCR and NIBRS errors
- * Don't waste time & resources maintaining multiple databases
- * Business processes must change to match new technologies
- * Ensure data integrity with a single point of entry; eliminate redundant data entry

For more information, visit the NJ web site at:
<http://www.ojp.usdoj.gov/nij/>



Best Practice Slick Sheet Version I Back

INFORMATION LED POLICING

TECHNOLOGY

O
O
L
S

"Develop technology solutions but do not fixate on them. Technological solutions to data management will certainly help but will not be able to address all of the concerns."

Jerry Ratcliff
*Information
Led
Policing*

It's not your father's law enforcement agency anymore.



Gone are the days of handwritten reports and the use of 3x5 index cards to store criminal histories. Today's model law enforcement agencies maintain wireless networks and manage multiple databases on varied platforms. Digital imaging across IP networks means that officers must now think in terms of gigabytes and terabytes. High-tech crime demands a high-tech response. Officers must have the ability to quickly gather, process, analyze, interpret, and distribute critical data necessary to apprehend criminals wherever they may reside.

Major Ronald D. Hairston, City of Danville, VA Police Department
2009 IACP LEIM Section Chairman

Best Practices

- Identify and involve all stakeholders early in the planning process
- Single point of entry for all data
- Include error-spotting capabilities in software where possible
- Eliminate redundant (old) databases
- Link system to GIS for address verification
- Visit agencies currently using systems under consideration-do not rely on a demo database
- Training-training-training include officers, investigators and records personnel
- Obtain source code for future customization

Best Practice Slick Sheet Version II Front

Quality Data can:

- Enhance safety for street officers
- Reduce identity errors
- Increase operational efficiency
- Foster inter-agency information sharing
- Provide timely & accurate information
- Improve customer service
- Identify crime hot-spots
- Boost effectiveness of law enforcement

**Data Quality is
Essential**

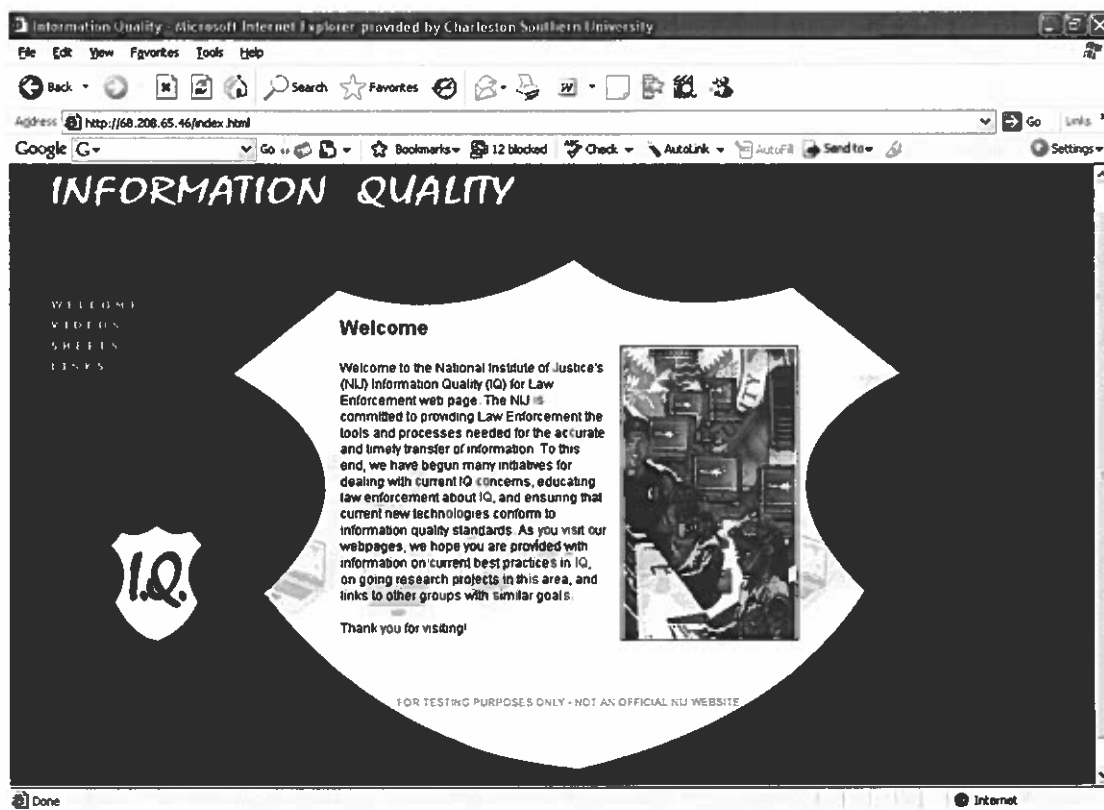
- * Ensure offenses are coded correctly in data field
- * All personnel must see the value of accurate data
- * Share your data—give so that you can receive!
- * Criminals don't follow jurisdictional lines; quality information can be shared among agencies and provide leads to solving multiple crimes
- * Strive for integration of all agency databases to eliminate duplicate data entry
- * Keep up with new technology—upgrade when possible
- * Make sure all users and view/print as needed
- * Develop records "specialists" and strive for continuity in the Division

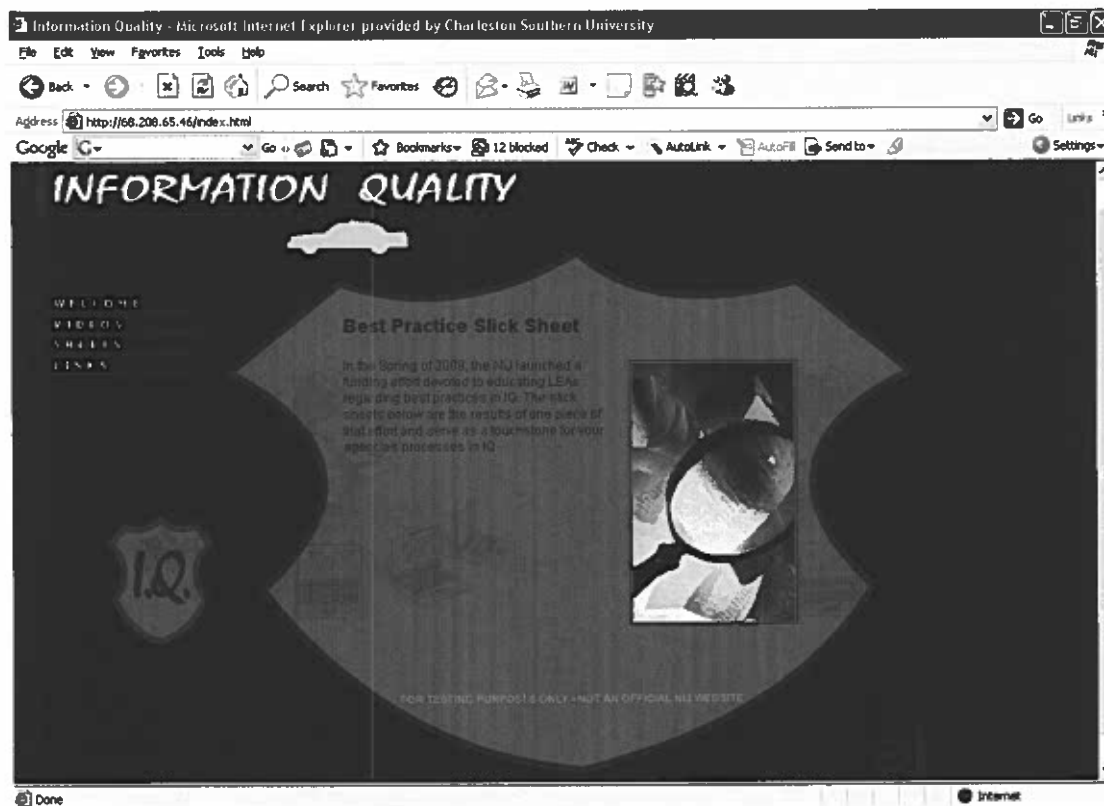
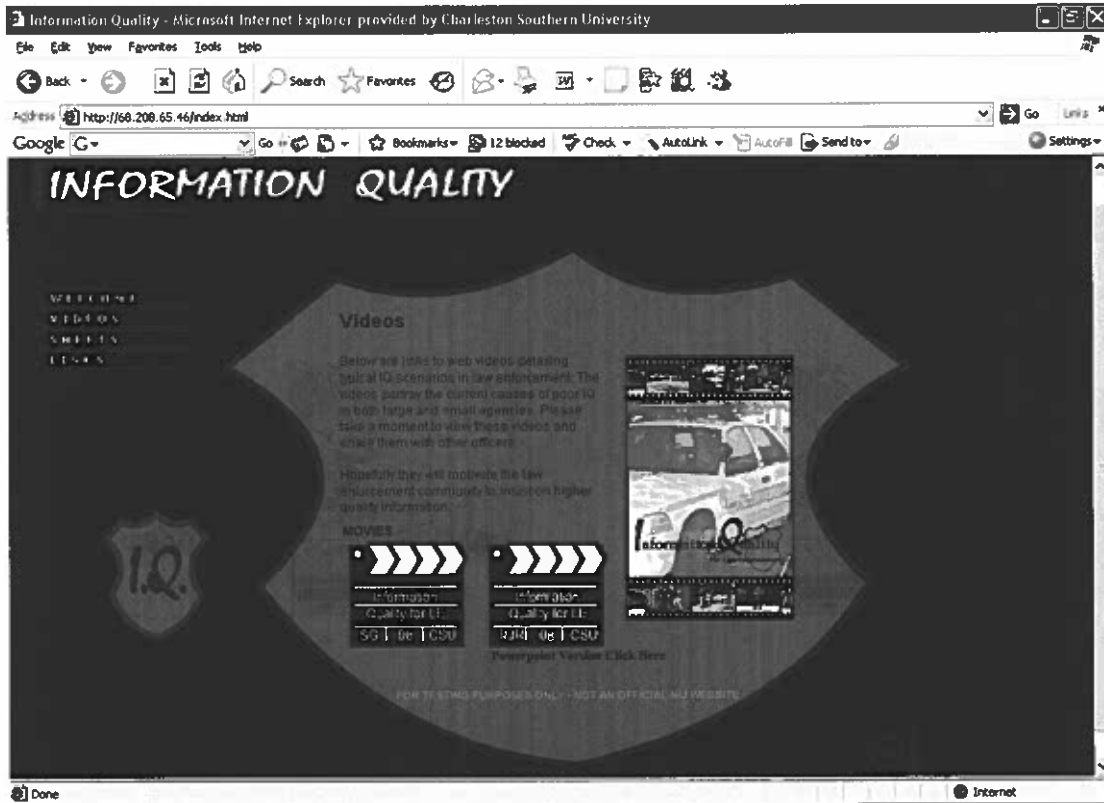
For more information, visit the NJ web site at:
<http://www.ojp.usdoj.gov/nj/>

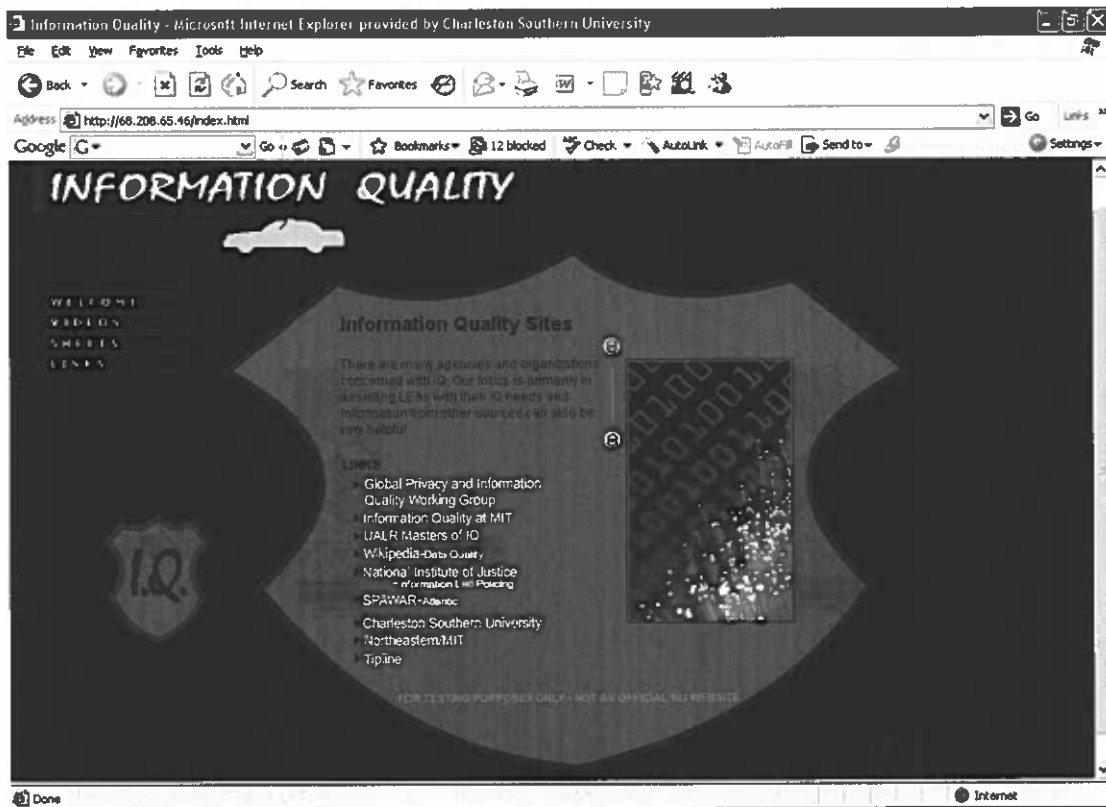


Best Practice Slick Sheet Version II Back

Appendix F Web Page Screen Shots

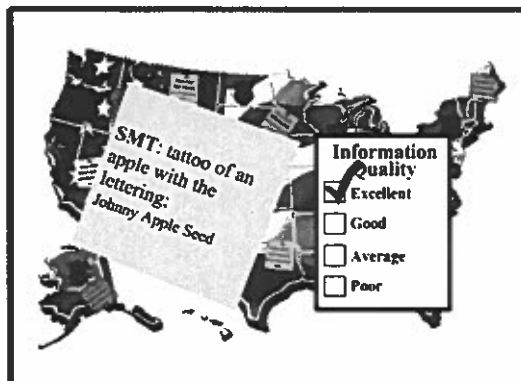






Appendix G Large Agency Video Powerpoint Presentation

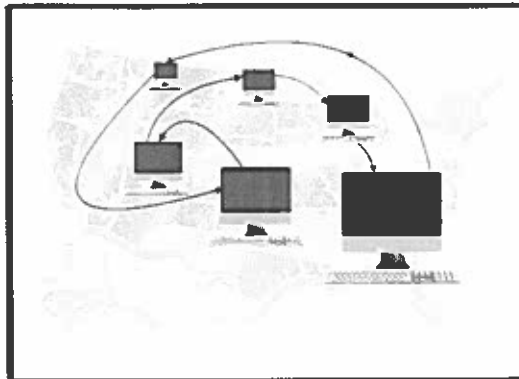
Information Quality for Law Enforcement



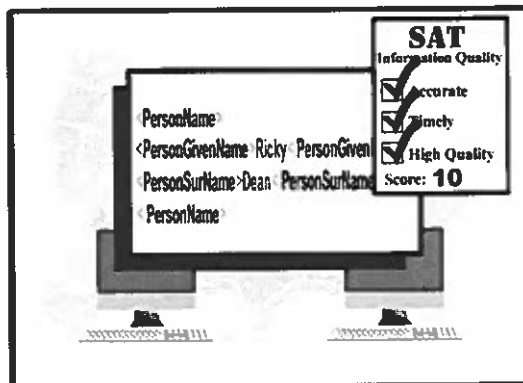
Law Enforcement agencies across the nation gather thousands of pieces of information daily. The need to share this information with other agencies is inevitable. But what about the quality of the information being shared? How reliable is it? Does it have the same meaning for one agency as it does for another?



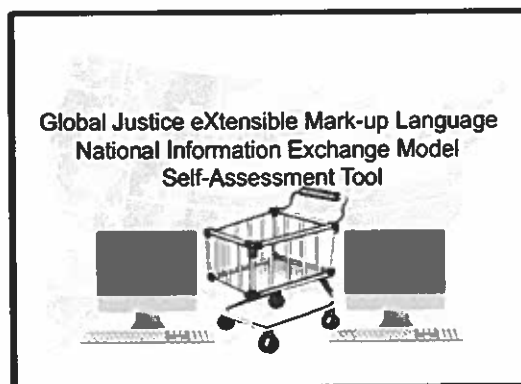
The Global Justice Extensible Mark-up language and National Information Exchange Model are XML standards that were developed by the National Institute of Justice and partnering agencies.



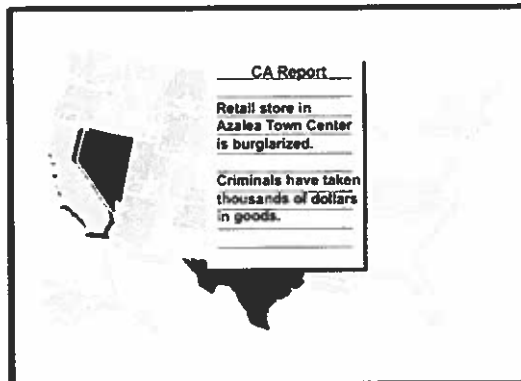
They were designed for the exchange of criminal justice information. This has facilitated new levels of information sharing between different agencies in a timely manner. This common standard eliminates the need for each agency to create its own exchange standards. With all systems using the same common terms, NIEM permits access from numerous sources and reuse in numerous applications.



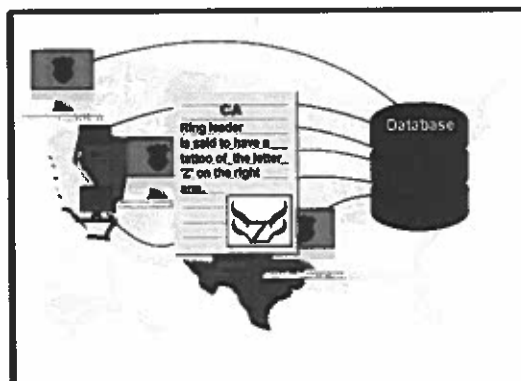
While using a common standard increases the timeliness and ability of agencies to share data, there is no measure of the quality of the information being exchanged. To increase the information quality and usability of the information, a self assessment tool is needed. This ensures that the data being shared is accurate, timely, and of overall high quality.



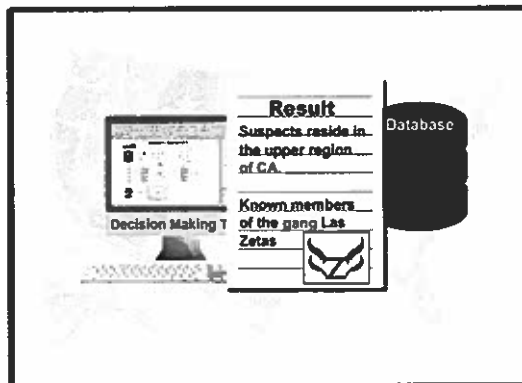
To illustrate how the GJXML/NIEM and the Self-Assessment tool enhance the information quality shared between different Law Enforcement agencies we will present a multi-State shoplifting scenario.



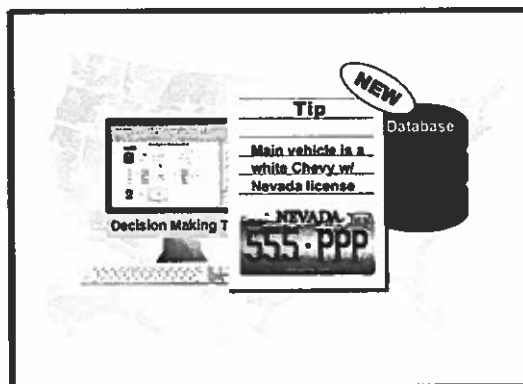
Shoplifting is a nationwide crime. Major retailers through out the States of Nevada, California and Texas have felt a strong increase in shoplifting activities. A retail store in California is burglarized and the criminals have taken thousands of dollars in goods. Hours later the Las Vegas Metropolitan police seize a van full of items from shoplifting episodes all over the southwest. The retail stores in Texas are also burglarized and the criminals are caught at the border attempting to cross over with the goods.



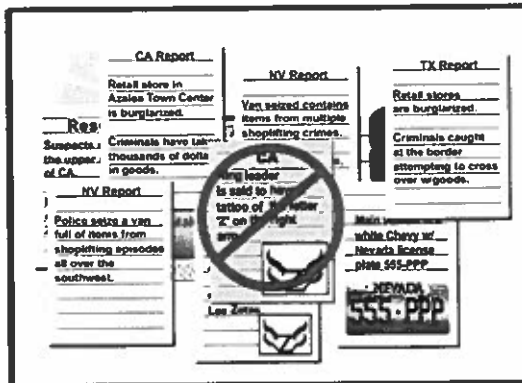
To track down the criminals, the retail stores and Law Enforcement have come together through a national online database. Each of them has entered their details into the database allowing Law Enforcement and other retailers to have almost real-time access to crime information as they occur. But the information entered into the database from the retailers in California is incorrect with regards to a gang tattoo marking. Law Enforcement believes these burglaries are linked to the same shoplifting ring.



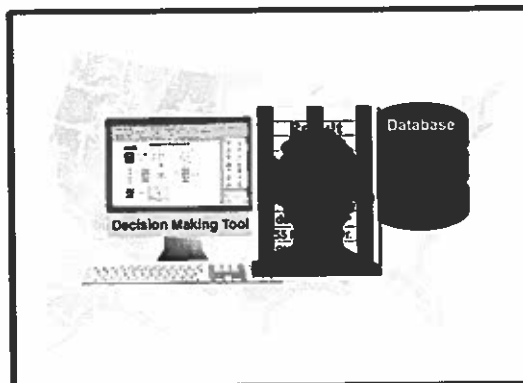
Once all the information is gathered from the retailers and Law Enforcement agencies, it's entered into a decision making tool. Based on the tattoo description, the decision making tool indicates that the gang known as Las Zetas in the upper region of California should be investigated for their possible involvement in the shoplifting crime.



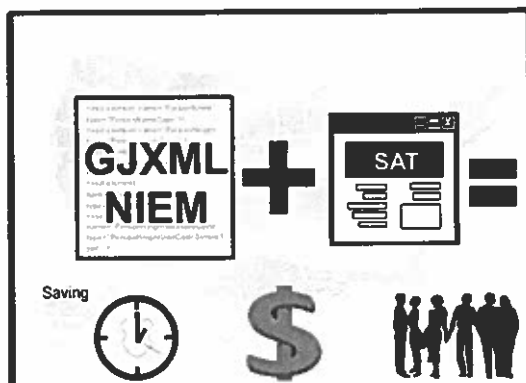
Days later a new tip is received from another Law Enforcement agency where the main vehicle in California is a white Chevy with Nevada license plate number 555-PPP. This new tip is entered into the decision making tool and the new report advises Law Enforcement to search the lower Los Angeles area for the same gang.



Law Enforcement begins to revise the information they have gathered thus far. They begin to notice that if they take out the SMT data, the decision making tool will advise them to search the Houston area where the leader of the Shoplifting Crime ring resides. With the new report, the correct Law Enforcement agency is alerted and the suspect is apprehended.



The erroneous piece of information was leading Law Enforcement to focus on the wrong area. Had the information been assessed through a self assessment tool, Law Enforcement would have been able to track down and catch the suspect sooner.



The GJXML/ NIEM standards and Self Assessment tools are two powerful tools developed to improve information quality for Law Enforcement agencies. Through their use, better decision making within Law Enforcement is possible. Through their use time, money and, more importantly, lives can be saved!

Appendix D.

Scripts.

IQ Station Narrator, Officer Valerie Voice over

- Scene: Voice over comes facing the impressive data station. The narrator gives a basic introduction to IQ and some description of the hardware.
- Shot: Full face of high tech looking display station.

NARRATOR:

(subject to review)

The information station is the heart of the information quality review. Several simple cross checks are performed on data at both entry and retrieval times so simple data errors can be caught. This can prevent some wasted time and money catching bad guys.

Lessons Learned with officer Valerie

Officer Valerie

(subject to review)

In this first scenario, Detectives Rikki and Lane are tapped to serve an arrest warrant, but there are a few complications... the stories to follow are true, but the names have been changed to protect the guilty.

SCENE 1 - Officers Rikki and Lane SERve warrants

- Rikki and Lane knock on the door of the suburban home. They are here to serve several arrest warrants for auto theft, larceny, assault, and fraud.
- The home is whitewashed and spotless, with a small fence and gate. It appears as if Ozzie and Harriet live here and one expects Doris Day to answer the door. There is more than likely a key under the potted plant by the door. The front door is Hunter Green and has white clear sidelights with obviously no alarm system or security. There is an American Flag on the left side of the door and a doorbell chime on the right. The white concrete side walk is neatly edged, and lined with 3 foot fluffy shrubs, that appear to your un-botanical mind to be azaleas.
- There is what appears to be a 'pink flamingo' in the front yard, completely ruining the ambiance and landscaped Feng-Shui perfection of the yard. It is obviously a male rebellion and clearly indicates criminal tendencies.
- The two detectives drive up in an unmarked cruiser with 62 antennae, 4 lights and a crash bar on the front.
- Frontal on the cruiser through the window.

DETECTIVE LANE

Here, this one.

RIKKI:

Are you sure?

LANE:

Yes, 129 JennyWhere Lane. This is it.

RIKKI:

What's that paperwork say again?

LANE:

Auto theft, larceny, assault, and fraud.

RIKKI:

That's crazy. Pharmacy hopping maybe.

RIKKI:

Will you look at that?

-She points to the bird.

LANE:

What?

RIKKI:

That ugly pink flamingo in the yard. It's completely out of place. Why not put a toilet next to it with flowers? I mean, come on.

LANE:

What's wrong with that, Rikki-Bobbi?

RIKKI:

Where are you from again?

LANE:

Mobile, Alabama.

RIKKI:

Well, that explains a lot.

- Lane puts the car in park and the two officers get out of the car, and come around to the front walk. They Stop just in front of the walk.

LANE:

Wait!

RIKKI:

What the heck?

LANE:

Where do we wipe our feet?

RIKKI:

Shut up!

- Lane and Rikki walk up to the steps and climb up to the porch. Lane reaches out and punches the doorbell chime. Somewhere in the house a set of chimes gently plays the first few bars of Dixie.

- Rikki and Lane look at each other in disbelief.

RIKKI:

I guess you never know, do you.

LANE:

That's really cool, I want one!

- Rikki shakes her head.

RIKKI:

Naturally.

- A young women (middle twenties) in an apron opens the front door and smiles prettily at the detectives.

Jane Doe

May I help you?

- Rikki pulls out the sheaf of warrants and makes some pretense of looking them over. She holds up her ID folder. Lane does as well.

RIKKI:

Ma'am, I am Detective Sergeant Alvarez and this is Detective Sergeant Goldberg. We are looking for a John Doe, Jr, do you know him?

JANE DOE

Well of course I do.

- Jane looks confused. Lane and Rikki look at each other again.

LANE:

May we speak with him, Ma'am?

- Jane looks even more confused, with overtones of shock.

JANE DOE

Of course, Officer. I'm not sure whether it will help you, but you certainly can.

- Jane turns around and yells into the house.

JANE DOE

Johnny, there are some policemen here to see you.

- There is a fairly significant pause, and then around the corner of the small foyer comes a little boy, around 3-4 years old carrying a blanket and sucking his thumb.

JOHNNY DOE

Policeman?

- Ricki and Lane look at each other again.

RIKKI:

Mrs. Doe, is Johnny's father home?

JANE DOE

No, he's at work, Officer.

DETECTIVE SARGEANT LANE

Where does he work, Ma'am

JANE DOE

In the Federal Building downtown, Officer. He works for the Dept of the Treasury.

RIKKI:

What does he do?

- Jane looks shocked and surprised again.

JANE DOE:

Frankly Officer, I thought you knew him. He works for Alcohol, Tobacco and Firearms.

- Rikki and Lane exchange another look.

RIKKI:

Well, ma'am, that is a very nice looking boy you have there. Have a nice day.

JANE DOE:

Of course, Officer.

- Lane is conspicuously silent. Rikki motions with her head and they head down the manicured walk. Jane switches expressions from shocked and confused to puzzled and relieved. She stands in the doorway to watch them leave. (Pan down the walk as the detectives leave.)

- Front of unmarked cruiser. They open the doors and get in with Lane driving. They

both speak at the same time.

RIKKI:

Don't say it.

LANE:

We've got warrants. We need to take him in.

- Voice over, narration about what actually happened:

NARRATOR:

Obviously Doe is not the family's name. There are 50 or so Doe warrants at any precinct at any time. The family had a common name, say Smith, for example. Somehow in the towns records, Jonny Jr's social security number was one digit off from the suspects with the same last name. At some time in the near past, the records had been switched. Johnny was enrolled in the program to record children's fingerprints, and thus his records were in the database.

Note: In the following scenario Bill Dooley is based upon an officer of my acquaintance named William Tunney. He has a lisp, and rather than being offended by it, he exploits it outrageously. Police being who they are, they find this hysterical.

Scene 2 - The Arrest

- Scene: Voice over comes facing the impressive data station. The narrator gives a basic introduction to IQ and some description of the hardware.

- A routine traffic stop leads to an invalid arrest wasting the officers time and the suspects. Invalid data in the system causes a warrant to be placed against the wrong license number. This is easily corrected using some very simple cross checking.

- Shot: Full face of high tech looking display station.

- Front of Cruiser: Officer Bill Dooley is enjoying a nice quiet evening, sitting on a back country road. He is enjoying quiet time in the cruiser, eating a sandwich and drinking coffee around doing paperwork. The cruiser lurches sideways as a family minivan passes by doing well over the speed limit. Radar readout reads a very high number. Officer Dooley has a speech impediment, which makes him difficult to understand under stress.

- SHot: in Through Cruiser Driver Side Window.

- Officer Bill is sitting in the drivers seat drinking his coffee. When

BILL:

Reapin' Rizzards!

- The radar alarm sounds and Bill jumps, spilling coffee over his freshly pressed shirt, not to mention the clipboard of citations. Bill pulls the cruiser out behind the van and flips on his lights and siren. After some moments delay he pulls in behind the green/silver minivan. The van pulls off on the shoulder, and the cruiser pulls in behind the van. Bill kills the siren, and calls Dispatch.

BILL:

128 to Station, be advised I wiw be making a twaffic top, about 2 my'yos up Twisty Wane fwom Bendew's End.

STATION:

Unit 128 Prease, Uh... PLEASE repeat.

BILL:

128 to Station, I shaid: BE advised I wiw be making a twaffic top, about 2 my'yos up Twisty Wane fwom Bendew's End.

STATION:

Woger, Unit 128

-Shot: Switch to cruiser cam.

- Bill does the police walk up on the drivers side of the little van, and knocks on the window. The young woman, Jeri Powell rolls down the window and smiles prettily

at him. She has her 2 year old daughter in the car.

BILL:

Can I see your Dwivers wicense and wegistwation pwease?

JERI POWELL:

Excuse me, officer?

BILL:

Why does evwybody make me wepeat myself? I said.. Can I see your Dwivers license and registration pwease?

- Jeri mostly successfully tries to hide a grin.

JERI:

Of course officer.

- Jeri hands her papers to Officer Bill and turns back to the little girl.

ISABELLA:

Mommy, why does that man talk funny?

JERI:

Hush!

ISABELLA:

But Mommy, he sounds wike Elmer Fudd!

JERI:

Hush!

- Bill takes the paper and does the walk back to the cruiser, gets in and runs the plates and registration on the terminal. After a small wait, the plates come back clean, but the license has an outstanding warrant for grand larceny, dated 2002. Jeri is 22. Born in 1986.

BILL:

128 to dispatch, this dwiver has an outstanding wawwant fowa gwand warceny. Pwease send me a femawe officeh fowah twanspoat. I'm packin' hew in!

Dispatch:

10-4. Dispatching Officer Lilly to your location.

BILL:

Woger dat, dispatch. Wiwwy is on the way.

- Bill gets out of the cruiser and does the police walk back up to the van.

BILL:

Ma'am, we have a wawwant fow your awwest, appawentwy, you faiwed to appeaw
in couwt, in 2002.

JERI:

Excuse me?

BILL:

I said... we have a wawwant fow your awwest, appawentwy, you faiwed to appeaw
in couwt, in 2002. Can you have someone meet you at the station to pick up your
daughtew? I have a femawe officer en woute.

JERI:

Officer, I don't understand!

- Jeri is getting visibly upset. Her daughter is starting to cry

ISABELLA:

Mommy is the bad man who talks funny going to take us away?

JERI:

Bella, I will call Daddy, and he'll get all this straightened out. Just be patient.

SHOT - STATION BOOKING.

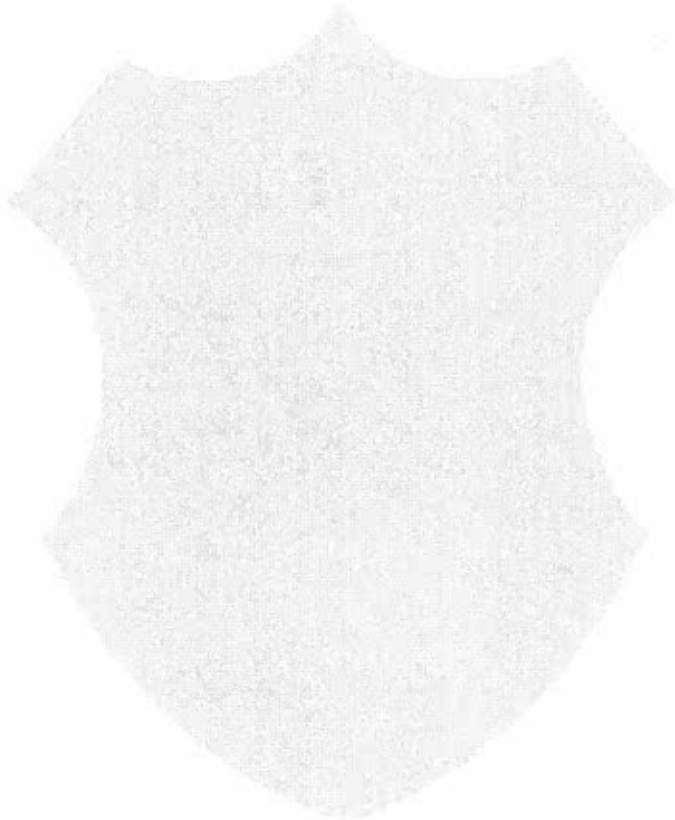
- VOICE OVER - Jeri Powell is arrested, Mirandized and transported to the station

for booking. The warrant is for grand larceny. In 2002, one Jerald Powell ordered furniture and electronics from a local rent-to-own and never returned the merchandise or paid for it. A complaint was filed against the license number given at the time of sale, and a summons was issued to the address given at the time of sale. Neither the license number nor the address match the particulars of Miss Powell's license.

(we can do the station scene here, where they discover the transposition and the charges against her are dropped.)

IQ Station Narrator, Officer Valerie Voice over

Appendix H. Final Defense Presentation Slides.



Information Quality/ Decision Making Scenario

Steven Garbeil, CSU
Rebecca Randall, CSU

Advisor
Dr. Valerie Sessions

1

Project Description

- **Educate LE about the importance of IQ**
 - Web streamed scenarios
 - Highlight nuances of poor IQ on LE and decision making tools
 - Convince LE of the need for high IQ
 - Point LE to resources currently available for use
 - Demonstrate future tools and implementations such as MIT/NU SAT

2

Decision Making Scenario

Headache causers

▪Contents

- Coming up with ideas to make the video
 - Low amount of imported images to keep small file size
 - Video size w/out audio 202KB
 - Video size w/audio (MP3) 711KB
- Matching the narration to the animation
 - Stream versus Start
- Preloader to start the video when loaded completely

7

Decision Making Scenario

Benefits

▪Contents

- Being a part of a research grant
- Learning about the process
 - Scripts
 - Audio
 - Filming
 - “Life Cycle Phase”

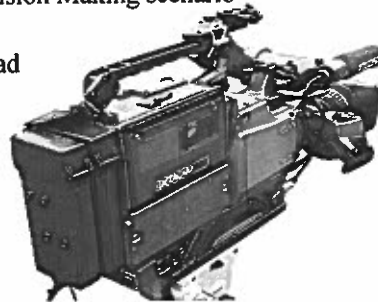
8

Decision Making Scenario

Live Scenario

■ Contents

- Web stream a few live IQ/Decision Making scenario
- Highlight how IQ problems lead to poor decisions



9

Fragments from the Film!

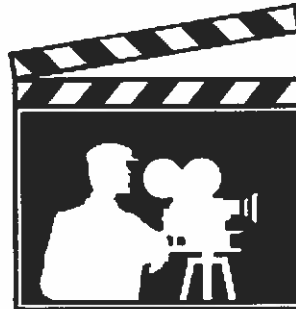


10



Video Production

- Pre-production
 - Brainstorming Sessions
 - Story Board
 - Scripting
 - Props and Set Design
- Production
 - Casting
 - Rehearsal
 - Production
- Post-Production
 - Editing
 - Packaging



11



Pre-production

- Brainstorming Sessions
- Story Board
- Scripting
- Props and Set Design



12



Brainstorming Sessions

- Getting a clear view of the project was initially difficult.
 - meetings with advisors
- Learning to read the Statement of Work for the Grant
 - what were we supposed to do?



13



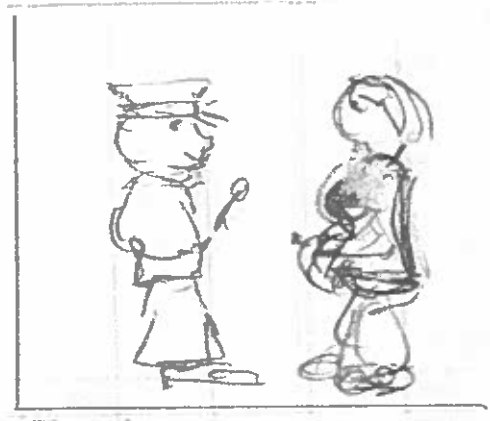
Notes about Video Production

- Video production is a highly computer intensive operation.
- Some schools have a special division of the computers science dept for video production
- Animation is extremely labor and time intensive, as well as challenging.
- Special Effect
 - Even simple special effects are difficult!

14



Story Board



- Officer "Bob" makes a traffic stop. The driver has 7 felony warrants for his arrest. The driver is a pregnant mother of 2.

15



Storyboard

- Created by Walt Disney for Snow White
 - Thumbnail sketches on a corkboard with tacks
 - Basic mechanics haven't changed for 100 years.
- Allows scenes to be represented and moved about.
 - first laid out in simple chronological order
 - a flashback, for example is simply a scene played after the resolution

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Storyboard

- Cool notes:
 - First film to be completely storyboarded was Gone with the wind
 - Video Games are now completely storyboarded.
 - Can be followed by animatic, or a photic.
 - Movie Storm is an animatic sequencer, now used to produce Sim 2 like animated movies
 - Hitchcock used optically correct storyboards to enforce his vision in North by Northwest.

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Scripting

- Script defines the production
 - In this case the script had to be written TO the scenarios which is the opposite of the normal process.
 - The specification of 'funny chuckle, not funny ha, ha'
 - Memorable but not offensive.
 - Scripting tools are very valuable.

18



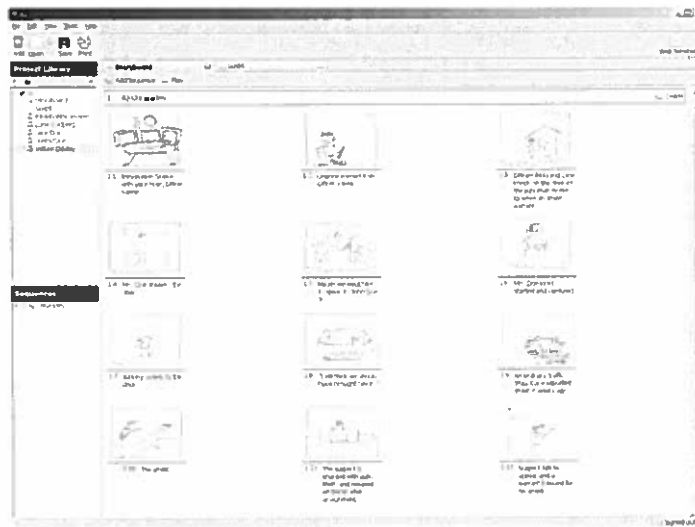
Scripting

- Dialog
 - A script consists of dialog, which defines the plot.
 - Actors read the dialog, and express it. These can be people (live film), Sims (mechanima) or cartoons.
 - Narration bridges gaps between dialog.
 - Sessions, tutorials, webcasts, trainings are all now scripted.

19



Celtx



20



Celtx



21



Props

- Theatrical Property
 - Anything held or used on stage to further the plot.
 - The flamingoes are .. Props ..
 - In the case of our extremely low budget production Props were very important.
 - Getting the cruiser was a huge success.



22



Production

- Casting
- Rehearsal
- Shooting/Filming
 - Done by Mark Kennedy, of Kennedy video.



23



Casting

- By Fiat.
 - The casting was done by decree.
 - Led to a small problem later.
 - Not able to control or use real actors.

24



Shooting

- Four Locations
 - Charleston Southern Inside
 - Charleston Southern Outside
 - Summerville PD (wasted!)
 - Summerville Downtown.
 - Mt. Pleasant Suburbia

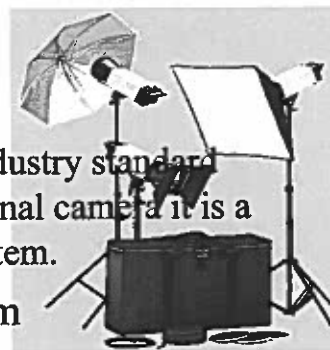


25



Shooting

- Sony Beta Camera
 - The Sony Betacam is the industry standard portable mid-level professional camera it is a digital/analog recording system.
- 8 track analog sound system
- Lighting
 - Screens and Masks



26



Post-Production

- Editing
- Packaging
- Delivery



27



Editing

- Editing was done with Adobe Premiere CS3
- Adobe Premiere is a non-linear digital editor.
- Video Editing is extremely time consuming and know-how intensive process.



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Packaging Video

- Editing is done in either uncompressed modes, or low compression
- MPEG2 is DVD's low compression.
- Uncompressed is approx a GB minute of video.
- Compressed on a Non Hi Def DVD, a movie is about ...90Min/4GB *2 for Hi-Def

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Packaging for Delivery

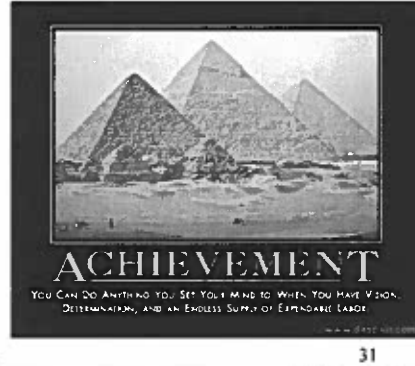
- Uncompressed video is a series of frames in a chronological order
- Compressed video is performed in several stages, with a keyframe and holding the differences.
- Then compressing the frames as data.

30



Challenges

- Plot
- Time Constraints
- Administrative Constraints
- Managing Actors
- Hardware
- Video format problems
- Scheduling Challenges



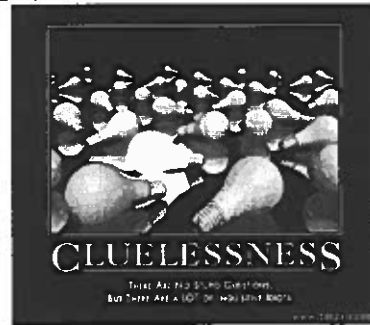
Lessons Learned *(SC Style)

- Not a project for a person with 'other' time demands
- Prior planning is extremely important.
- Layout video and sound formats in the scripting.
- Actors need more rehearsal
 - especially instructor type actors!



Questions!

- Any questions can be addressed at this time
- If any of the answers require research we will email you the reply (... eventually.)



CHARLESTON SOUTHERN UNIVERSITY

Computer and Information Sciences